Team Handbook

Lung-Cancer Risk Data Science Project

The single source of truth: verified data findings, CRISP-DM roadmap, concepts to master, task cards for seven people, report and presentation blueprints, reproducibility rules and a landscape of existing work.

Midterm presentation: Wednesday, 28 October 2026 (20 minutes)
Assessment: Data Science Midterm Project (code 40, presentation 35, report 25)
Scope: CRISP-DM phases 1-6 of the rubric, with modelling previewed as future work

Internal team document | Version 1.0 | Compiled Saturday, 24 October 2026 | Midterm: Wednesday, 28 October 2026

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How to use this handbook

This document replaces scattered chat history. If something is not written here, it is not decided. Every claim carries a status tag so that nobody has to guess whether a statement was measured, sourced, hoped for or merely assumed.

Tag	Meaning	Rule for using it in the report or slides
[VERIFIED]	Computed by running code on the actual files on 24 Oct 2026.	Safe to quote after you re-run the repo and your own number matches.
[SOURCED]	Taken from an external source listed in the References chapter.	Cite the source; paraphrase, never copy long passages.
[DECISION]	A choice the team has adopted (or must adopt tonight).	Follow it; changes go through the decision log.
[PLANNED]	Intended work that has not been done yet.	Never present as completed. Slides must label planned work as planned.
[ASSUMPTION]	Believed but not checked.	Convert to VERIFIED or OPEN before Monday.
[OPEN]	A question that only the instructor or the data provider can answer.	Owner A resolves; until then use the conservative interpretation.
[JUDGMENT]	Expert opinion of the handbook author.	Discuss it; you may overrule it with evidence.

Reading paths

If you are...	Read first	Then
Everyone (tonight, 30 min)	Ch. 1 Executive summary, Ch. 2 Requirements, Ch. 4 Verified findings	Ch. 9 Team operating system, your task card in Ch. 10
A - Team lead	Everything; Ch. 5 Decisions, Ch. 13 Presentation	Appendix E open questions
B - Data understanding	Ch. 3 Domain primer, Ch. 6 phases 1-2	Ch. 12 Report blueprint section 2
C and D - Audit and cleaning	Ch. 4, Ch. 6 phase 3, Appendix B code	Ch. 8 Reproducibility
E and F - EDA and statistics	Ch. 6 phase 4, Ch. 7 statistics primer	Ch. 13 Q and A bank
G - Features and report assembly	Ch. 6 phase 5, Ch. 12	Ch. 14 Future modelling plan

What this handbook is not

It is not a clinical document. The project is a research prototype for coursework. Nothing here supports a diagnostic, screening or treatment claim, and no slide, report sentence or answer in Q and A may imply one. It also does not replace running the code: every number marked VERIFIED must be regenerated by the team from the repository.

Provenance of this document. Compiled on Saturday 24 October 2026 from (1) the two attached files: the ZIP *Lung-cancer-early-detection-main.zip* and the rubric PDF *Data_Science_Midterm_Project_Rubric.pdf*; (2) code executed on the ZIP contents (Python 3.12, pandas 3.0.2, numpy 2.4.4, scipy 1.17.1, statsmodels 0.15.0, scikit-learn 1.8.0); (3) web research listed in the References chapter. The raw CSV was read directly from the ZIP and never modified.

1. Executive summary

1.1 Where we stand

- **Deadline.** Midterm presentation on Wednesday 28 October 2026: 20 minutes, graded together with executable code and a written report. Today is Saturday 24 October, so there are four working days. [VERIFIED]
- **What we actually have.** One data file (*lung_cancer_combined_22457.csv*, 22,457 rows, 17 columns) and a README that contains only a title. There is no data dictionary, no source link, no licence and no second dataset. [VERIFIED]
- **What was assumed earlier and is wrong.** The brief spoke of several datasets and of 'early detection'. There is one file, and its columns are self-reported symptoms and habits, not imaging, biomarkers or smoking dose. 'Early detection' is therefore not a claim this data can support. [VERIFIED]
- **Rubric.** Model training is explicitly not required for the midterm. Points go to data diagnosis, defensible cleaning, leakage prevention, EDA with a justified hypothesis test, feature engineering, and the quality of explanation. [SOURCED]

1.2 The headline finding (read this twice)

> **The apparent signal in the raw file is produced by data-integrity artefacts, not by patients**
>
> In the raw file all 14 binary features are 'significantly' associated with the label (Cramer's V up to 0.144; smoking odds ratio 1.49, p = 4.5e-30). After removing exact duplicates and the artefact block described below, every association collapses: smoking odds ratio 1.00 (95% CI 0.92 to 1.08), largest Cramer's V 0.015, no feature survives multiple-comparison correction (smallest BH q = 0.46). A shallow model on the raw file reaches AUC about 0.67 to 0.70; on the de-duplicated file it falls to 0.50 (chance). [VERIFIED]

- **Artefact 1 - an ordered, duplicated, all-negative block.** Rows 1 to 1,455 (patient_id 1 to 1,455) are 100% negative but contain only 103 distinct feature patterns, repeated 2 to 86 times. Their symptom prevalence differs sharply from every later row (e.g. smokers 29% vs 56%; GENDER code 1 in 18% vs 55%). The first positive label appears at row 1,456. [VERIFIED]
- **Artefact 2 - exact duplicates and label conflicts.** 2,573 rows duplicate another row exactly (features and label). 133 feature patterns carry both labels (315 rows). [VERIFIED]
- **Artefact 3 - the identifier leaks.** patient_id alone separates the classes with AUC 0.67 because of the ordering. It must never be a feature. [VERIFIED]
- **Beyond row 1,455 the data behave like independent coin flips.** After de-duplication, symptom columns are essentially uncorrelated (max |r| = 0.022; 4.4% of 91 pairwise tests significant at 5%, which is the chance rate), AGE is almost exactly bell-shaped (skewness -0.01), a PCA has no dominant component, and positives and negatives have the same feature profile (e.g. smokers 56.0% vs 56.2%). This is consistent with programmatic generation or resampling; it does not prove it. It is also unlike real self-reported clinical data, where smoking, age, chronic disease and symptoms are correlated. [VERIFIED]

1.3 What this changes about our strategy

A team that runs a classifier on the raw file will report 'good' accuracy that is an artefact (82% of rows are positive; shuffled cross-validation with duplicates inflates results further). We do the opposite and win on exactly the dimensions the rubric rewards: **diagnose, justify, prevent leakage, quantify sensitivity, interpret statistics honestly**. This is the **integrity-first strategy** [DECISION]. It turns an awkward dataset into a rigorous, memorable midterm and produces a reusable audit pipeline for the rest of the semester.

> **Seven decisions to confirm tonight (details in Ch. 5)**
>
> D-01 raw data immutable and checksummed. D-02 integrity-first framing. D-03 primary analysis view = exact duplicates removed (view B); C and D are pre-specified sensitivity analyses; A is shown only to quantify the artefact. D-04 patient_id is excluded from every feature set. D-05 no model training in the midterm deliverable. D-06 GENDER is reported as a code (0/1) until its meaning is verified. D-07 one repository, one Makefile, seed 42, every number regenerated by code.

1.4 Critical path and top risks

Risk	Why it matters	Mitigation (owner)
Only four days	Seven people, one dataset, three graded artefacts.	Fixed task cards, daily integration at 21:00, freeze Tuesday 23:59 (A).
Finding challenged: 'you removed data'	Instructor may expect a normal pipeline on the whole file.	Show all four views side by side; the conclusion is the same in B, C and D; keep A in the deck (C, D, A).
Members cannot answer questions	8 pts for Q and A, and the rubric says all members must be ready.	Cross-training pairs, 35-question bank in Ch. 13, two mock Q and A rounds (A).
Numbers drift between code, report and slides	Rubric scores consistency across the three.	One results.json from code; slides and report quote only from it (G).
Overclaiming clinically	Credibility loss; ethical issue.	Banned-phrases list in Ch. 12; every slide footer says research prototype (A).

2. Official requirements and compliance

Source: *Data_Science_Midterm_Project_Rubric.pdf* (attached). It defines the assessment scope as CRISP-DM from Business Understanding through Feature Engineering; model training and evaluation are not required. **[SOURCED]**

2.1 Scoring structure

Component	Points	Sub-criteria (points)
Technical implementation and code	40	Data loading and organization (5), Cleaning and preprocessing (12), EDA and statistical analysis (13), Feature engineering (7), Code quality and reproducibility (3)
Oral presentation and discussion	35	Structure and clarity (6), Methodological explanation (9), Results and interpretation (9), Technical understanding and Q and A (8), Delivery and time (3)
Written project report	25	Business understanding (4), Data description and sources (4), Preprocessing methodology (5), EDA and statistical interpretation (6), Feature engineering (3), Conclusions, organisation and references (3)
Total	100	Points reward correctness, completeness, justification, interpretation and consistency across code, report and presentation.

Phase content required by the rubric. (1) Business understanding: context, problem, focused research questions, measurable success criteria. (2) Data understanding: sources, rows, columns, types, units, key variables, initial quality issues. (3) Preprocessing: missing values, duplicates, inconsistent types or categories, outliers; clean each source and explain integration if there are several; **prevent leakage in any transformation**. (4) EDA: descriptive statistics, plots, correlation analysis or heatmaps when appropriate, relationships between variables, **at least one justified statistical hypothesis test with interpretation**. (5) Feature engineering: selection, extraction or creation, encoding, scaling or transformation; justify the final feature set. (6) Conclusions: evidence-based findings, revisit the questions, limitations, reasonable future modelling steps **without implementing models**.

Submission package. Code (runnable notebook(s) or scripts, dependencies and instructions, access to the data or a documented source); report (structured, six phases, figures and tables with captions, interpretation, references); presentation (problem, data, preprocessing, key EDA, engineered features, conclusions, limitations; all members ready for technical questions).

2.2 Compliance matrix: every rubric line has an owner and evidence

Rubric line (pts)	Our deliverable	Owner	Evidence file
Data loading and organization (5)	Checksum-verified loader, schema validation with asserts, data dictionary	B	src/data_io.py, docs/data_dictionary.md
Cleaning and preprocessing (12)	Audit of missingness, duplicates, conflicts, outliers, types, order; four views A-D; before/after log; leakage rules	C, D	src/audit.py, src/clean.py, reports/tables/cleaning_log.csv
EDA and statistics (13)	Descriptives, 7 core figures, correlation, 14-feature chi-square family with BH, H1-H5 with effect sizes and CIs	E, F	notebooks/03_eda.ipynb, reports/figures/
Feature engineering (7)	Written feature spec, three feature sets F0/F1/F2, verification asserts, selection rule	G	src/features.py, docs/feature_spec.md
Code quality, reproducibility (3)	Makefile, pinned requirements, seed, pytest, no absolute paths	A, D	Makefile, requirements.txt, tests/
Structure and clarity (6)	Single storyline: question, audit, cleaning, evidence, features, limits	A	slides/outline in Ch. 13
Methodological explanation (9)	Each slide states why the method was chosen and its assumptions	all	speaker notes
Results and interpretation (9)	Seven figures, each with one takeaway sentence	E, F	reports/figures/
Technical understanding and Q and A (8)	Cross-training pairs, 35-question bank, two mock rounds	all	Ch. 13
Delivery and time (3)	19:30 target, two timed rehearsals	A	rehearsal log
Report: six sections (25)	Blueprint with word budgets and figure list	G + owners	Ch. 12

2.3 Requirements we could not verify

- Report and code **submission deadline and format** (file type, page limit, language) **[OPEN]**. Conservative assumption: due before the presentation; report as PDF of 10 to 14 pages; English.
- Length of the **Q and A** after the 20 minutes and whether it is per member **[OPEN]**. Assumption: 5 to 10 minutes, questions directed at individuals.
- Whether the **dataset was assigned** or chosen by the team, and whether the instructor expects modelling on it later **[OPEN]**.
- **Grading rubric for 'competition'**: the brief says the scope is competitive; the rubric file shows no ranking rule **[OPEN]**. We therefore optimise rubric points first and distinctiveness second.

2b. Contradictions found and how they are resolved

A good project lead resolves inconsistencies explicitly instead of carrying them silently. Each row below states the evidence and the adopted resolution.

#	Contradiction	Evidence	Resolution
X1	Brief says several raw datasets; the ZIP has one CSV.	ZIP listing: README.md (30 bytes: only a title) and one CSV.	Treat as single-source project. The 'integration' rubric item is satisfied by documenting that the file is already a combination and by auditing its lineage (Ch. 4).
X2	Repository name says 'early detection'; the columns are self-reported symptoms and habits.	17 columns; no imaging, no biomarkers, no smoking dose or duration, no time variable.	Frame as 'symptom-based risk indication on a survey file'. Never claim early detection (USPSTF screening is defined by age, pack-years and quit time).
X3	Earlier message said the signal is 'weak' (Cramer's V max 0.144, tree AUC about 0.66).	Full sensitivity analysis shows the signal is absent after removing duplicates and the leading block (V max 0.015).	Superseded. Correct statement: the raw-file signal is an artefact.

#	Contradiction	Evidence	Resolution
X4	Brief wants a 'technically advanced' project; the rubric forbids model training for the midterm.	Rubric: 'without implementing models'.	Advanced = rigour, not model count: audit, sensitivity analysis, FDR, effect sizes, reproducible tests. Models only as planned work (Ch. 14).
X5	Public documentation disagrees with itself.	Kaggle-derived pages: '284 instances' but a 309-row table; one project says the data are balanced, another that negatives outnumber positives; another dataset page describes vital signs and smoking categories that its columns do not contain.	Trust only what we compute. The original 309-row file was not retrievable in this environment; obtaining it is task B-4 for the lineage check.
X6	GENDER is coded 0/1 while all symptoms are coded 1/2.	Schema check.	Recode symptoms to 0/1 (2 = YES) and keep GENDER as GENDER_CODE until the source confirms the mapping.
X7	One public project drops AGE for 'high multicollinearity'.	Our VIF for AGE is 1.00 (all features 1.00 to 1.03).	Do not drop AGE for that reason. Dropping needs evidence from our own data.
X8	The brief gives 'Data Preparation' as one phase; the rubric splits preprocessing, EDA and feature engineering.	Rubric phases 3, 4, 5.	Follow the rubric numbering in all artefacts; map to CRISP-DM in Ch. 6.

2b.1 Strategy: how we outperform other teams

The following is a [JUDGMENT]: other teams are unknown, so we compare against the common pattern in public notebooks on this kind of survey data.

Typical pattern (assumed)	What we do instead	Rubric lines it earns
Load CSV, show head and describe, draw a heatmap.	Checksum, schema asserts, row-order and source-shift tests, before/after counts for every step.	Loading (5), Cleaning (12)
Drop duplicates silently, or never check.	Quantify duplicates, label conflicts, multiplicities; compare four policies.	Cleaning (12), Methodology (9)
Report many significant p-values (n is huge).	Effect sizes, CIs, BH correction, power and minimum detectable effect, permutation check.	EDA and statistics (13)
Chase accuracy with a classifier (82% is the majority baseline).	Explain why accuracy is meaningless here; plan leakage-safe evaluation; no premature models.	Q and A (8), Conclusions (3)
Slides full of default plots.	Seven purposeful figures, each answering one stated question, with one takeaway line.	Results (9), Clarity (6)
Code in one notebook with absolute paths.	Package with Makefile, tests and pinned environment; clean-clone run in under five minutes.	Code quality (3), Reproducibility
No limitations section, or generic ones.	Specific limitations tied to evidence: provenance, synthetic-like structure, construct validity, prevalence 82-87%.	Report (3), Q and A

3. Problem and domain primer

3.1 Clinical context [SOURCED]

- Lung cancer is the leading cause of cancer death in the United States. For 2020 an estimated 228,820 people were diagnosed and 135,720 died; overall five-year survival is 20.5%, while early-stage disease has a better prognosis and is more amenable to treatment (USPSTF, JAMA 2021; summarised by Applied Radiology).
- Smoking is the most important risk factor and increasing age is the second. The **USPSTF 2021** recommendation (grade B) is annual low-dose CT for adults aged 50 to 80 with at least 20 pack-years who currently smoke or quit within the past 15 years. It replaced the 2013 rule (55 to 80 years, 30 pack-years).
- In the National Lung Screening Trial (NLST), low-dose CT lowered lung-cancer mortality by roughly 15 to 20% relative to chest X-ray; more than 25% of screened people screened positive but fewer than 10% of those positives had or developed cancer (NLST analysis, PMC10002794). False positives are a central cost of screening.
- Validated risk models use **dose-like** smoking variables. PLCOm2012 (Tammemagi et al., NEJM 2013; 80,375 ever-smokers) uses, among others, age, education, BMI, family history, COPD, smoking status, intensity, duration and time since quitting, and was more sensitive than the NLST criteria (41.3% fewer lung cancers missed). A yes/no 'SMOKING' flag cannot reproduce that information.

> **Why this matters for our framing**
>
> Screening eligibility is defined by age, pack-years and quit time. Our file has none of the last two. Any sentence of the form 'our model detects lung cancer early' would therefore be unsupported. The defensible scope is: *examine whether self-reported symptom and lifestyle flags in this file carry reproducible information about the recorded label, and build the analysis foundation that a leakage-safe study would need.*

3.2 What does the label mean? [OPEN]

The column LUNG_CANCER is coded 0/1. Public descriptions of the related survey dataset say the data were collected from a website called an 'online lung cancer prediction system', and one paper describes the label as whether a person 'is suspected of having lung cancer based on symptoms and lifestyle'. If the label came from a website's own scoring rule, the dataset would measure that rule, not disease. This is a **construct-validity** risk: the target may not be a clinical diagnosis. Until the provenance is established we call it 'the recorded label' and never 'diagnosis' ([DECISION] D-09).

3.3 Dataset lineage

Item	What public pages say [SOURCED]	What we verified [VERIFIED]
Kaggle 'Lung Cancer' survey (user mysarahmadbhat)	16 attributes; description lists 284 instances while the table has 309 rows; YES=2, NO=1; collected from an online lung cancer prediction system; Gender M/F.	Our file uses the same 15 feature names and 1/2 coding for symptoms, but GENDER is 0/1 and the label is 0/1.
Kaggle 'Lung Cancer Dataset' (user akashnath29)	3,000 rows with the same symptom columns; prose describes age at diagnosis, smoking categories and vital signs that the columns do not contain.	Not available offline; relation to our file unknown.
Our file lung_cancer_combined_22457.csv	Nothing: README contains only the repository title.	22,457 x 17; adds patient_id (sequential 1 to 22,457); contains an ordered duplicated block and independent-looking bulk (Ch. 4). Provenance unknown.

Task B-4: obtain the original 309-row file and the 3,000-row file from Kaggle (needs a Kaggle login; the sandbox used for this handbook could not download them), then test whether their rows appear in our file. If the 103 leading patterns match the original survey, the leading block is probably resampled real survey rows and the bulk is generated; if nothing matches, the file is probably wholly generated. Either result is reportable.

3.4 Ethics and responsible use

- **No personal identifiers** besides a sequential id; still treat the data as health-related and do not publish it without checking its licence ([OPEN]).

- **Research prototype**: slide footers and the report abstract must state that nothing here is a validated diagnostic tool.
- **Representativeness**: age 21 to 87 (mean 62), prevalence 82% in the raw file and 87% after cleaning, versus a screening population in which most people do not have cancer. Prevalence-dependent metrics (accuracy, PPV) would be meaningless.
- **Fairness**: with GENDER code unverified and no ethnicity, subgroup analysis is limited to GENDER_CODE and age bands; say so.
- **Reporting standard**: for later modelling we adopt the TRIPOD+AI 27-item checklist (BMJ 2024) as the reporting guide, and the leakage 'model info sheet' idea of Kapoor and Narayanan (Patterns 2023).

4. Verified data findings (Data Understanding and audit)

Everything in this chapter was produced by the code in Appendix B and the script in Appendix A, run on 24 October 2026 on the CSV read directly from the ZIP. Re-run it: `make all`. If your numbers differ, stop and tell A immediately. **[VERIFIED]**

4.1 Inventory and data dictionary

Single file *lung_cancer_combined_22457.csv*, 864,917 bytes, 22,458 lines including the header, ASCII CSV, SHA-256 prefix `4ad2333067fcb0a9`. 22,457 rows x 17 columns, all int64, no missing values, no sentinel codes (symptoms contain only 1 and 2; AGE has no impossible values).

Column	Type / coding	Overall share = 2 (YES)	Notes
patient_id	identifier	-	1 to 22,457, sequential, unique. Row order carries label information: never a feature.
GENDER	0/1 code	50.7% code 1	Mapping to sex unverified. Original survey used M/F.
AGE	years, integer	mean 62.08, sd 8.28	21 to 87; bell-shaped bulk; 221 rows outside the IQR fences 40.5 to 84.5.
SMOKING	binary, 1 = NO, 2 = YES	54.5%	current smoking (self-report); no missing values
YELLOW_FINGERS	binary, 1 = NO, 2 = YES	54.2%	yellow fingers (nicotine staining proxy); no missing values
ANXIETY	binary, 1 = NO, 2 = YES	48.0%	anxiety; no missing values
PEER_PRESSURE	binary, 1 = NO, 2 = YES	47.7%	peer pressure; no missing values
CHRONIC_DISEASE	binary, 1 = NO, 2 = YES	48.7%	chronic disease; no missing values
FATIGUE	binary, 1 = NO, 2 = YES	66.1%	fatigue; no missing values
ALLERGY	binary, 1 = NO, 2 = YES	52.8%	allergy; no missing values
WHEEZING	binary, 1 = NO, 2 = YES	52.3%	wheezing; no missing values
ALCOHOL_CONSUMING	binary, 1 = NO, 2 = YES	52.3%	alcohol consumption; no missing values
COUGHING	binary, 1 = NO, 2 = YES	54.9%	coughing; no missing values
SHORTNESS_OF_BREATH	binary, 1 = NO, 2 = YES	63.0%	shortness of breath; no missing values
SWALLOWING_DIFFICULTY	binary, 1 = NO, 2 = YES	44.4%	difficulty swallowing; no missing values
CHEST_PAIN	binary, 1 = NO, 2 = YES	52.6%	chest pain; no missing values
LUNG_CANCER	0/1 label	82.1% positive	1 = recorded positive. Meaning of the label unverified (Ch. 3.2).

4.2 Target and age at a glance

Positives 18,434 (82.09%); negatives 4,023. A classifier that always answers 'positive' scores 82.1% accuracy, so accuracy is not an informative metric (Ch. 14). AGE: median 62, interquartile range 57 to 68, only 123 rows under age 40.

4.3 Artefact 1: the file is ordered

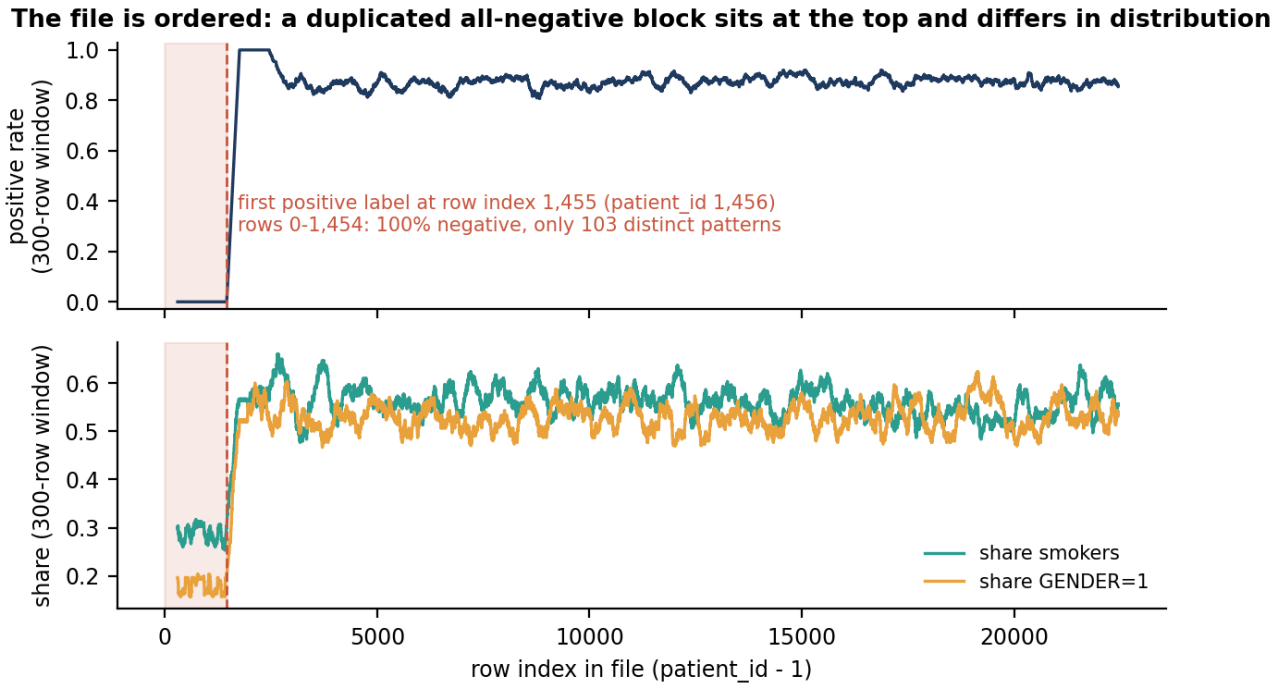


Figure 1. Rolling positive rate and rolling feature shares by row position. Rows 0 to 1,454 (0-based; patient_id 1 to 1,455) are all negative and differ in distribution; the first positive is at 0-based index 1,455 (patient_id 1,456). Source: audit.py on the raw file.

- The leading block contains **103 distinct feature patterns** in 1,455 rows (each repeated between 2 and 86 times, median 9). All 103 patterns have appeared by row 247; only 3 later rows match any of them.
- Compared with later negatives its symptom prevalence is far lower (Figure 2) and its AGE range reaches 21 (the rest of the file starts at 30); Cramer's V between block and later negatives is 0.18 to 0.54 per feature with all p below 1e-25. The block is not exchangeable with the rest of the file: it behaves like a separate source.
- **Consequence.** patient_id alone gives AUC 0.67 against the label. Any unshuffled split, and any model that sees the identifier or the row position, learns the artefact.

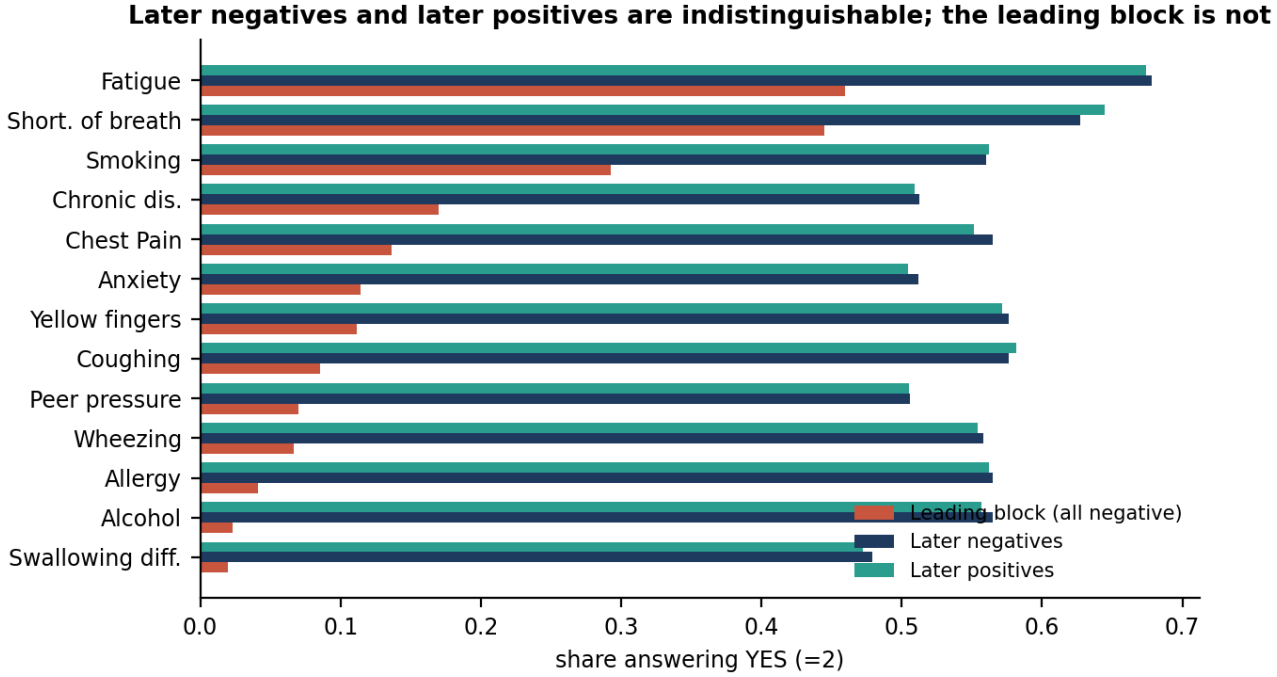


Figure 2. Share answering YES per symptom: leading block vs later negatives vs later positives. The two later groups are almost identical; the leading block is the outlier.

4.4 Artefact 2: duplicates and label conflicts

Quantity	Value	Meaning
Rows duplicating an earlier row (features and label)	2,573	1,352 sit in the leading block (1,455 rows reduce to 103), 1,221 in the later rows.
Rows duplicating on features only	2,706	133 further rows repeat a feature pattern with the opposite label.
Distinct feature patterns	19,751	865 patterns occur more than once; maximum multiplicity 86.
Patterns that carry both labels	133	315 rows. Label noise or a labelling rule that is not a function of these features.
Duplicates among rows 1,455 onward vs chance	1,352 vs about 600	Expected count if each column were an independent draw from its own marginal distribution (simulation, 10 replicates). Observed is roughly double, so a portion of later rows are copies.

Why duplicates are harmful and why 'maybe they are real people' is not enough

Fourteen binary answers and an integer age give about one million possible patterns, so chance collisions among 22,457 rows are expected to be rare, and are about 600 here. Exact copies of a row inflate n (so p-values shrink, 'pseudo-replication'), shift class proportions, and, worst, put the same row in both training and test folds ('duplicate leakage', one of the eight leakage types in the Kapoor and Narayanan taxonomy). Even if some duplicates were real people, we cannot tell which, so the transparent policy is: remove exact copies, and show that conclusions do not depend on that choice.

4.5 Artefact 3: after row 1,455 the columns behave like independent draws

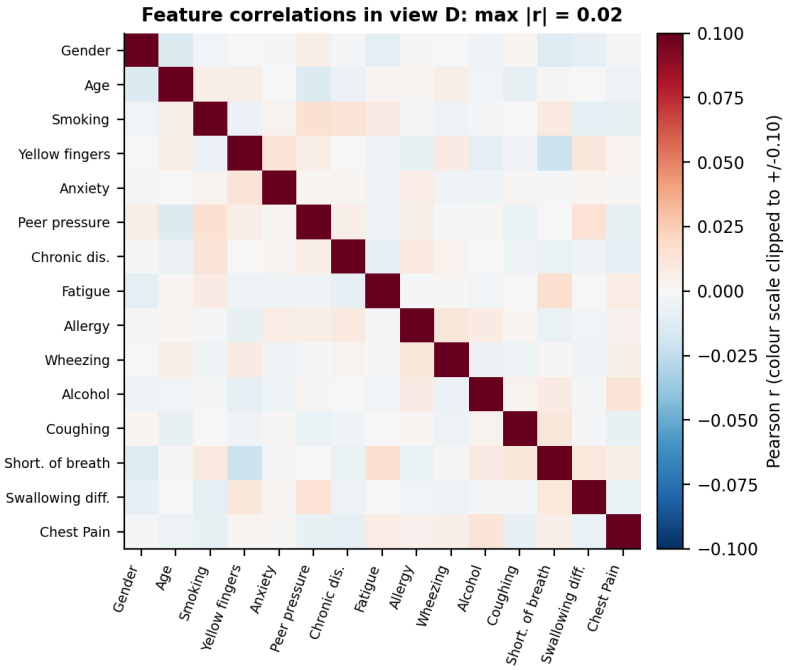


Figure 3. Pearson (phi) correlations among all 15 features in view D. Colour scale clipped to +/-0.10; the largest absolute off-diagonal value is 0.022.

- Pairwise chi-square tests among the 14 binary features (91 pairs): 4.4% reject at 5%, which is exactly what independence predicts. A PCA on standardised features has a first component explaining 7.0% of variance against 6.7% for a perfectly flat spectrum. VIF is 1.00 to 1.03 for every feature.
- Later positives and later negatives have the same profile on every feature (Figure 2); the positive rate is stable around 87% across the file.
- AGE in rows 1,455 onward: mean 62.24, sd 8.20, skewness -0.006, excess kurtosis -0.055 (Figure 4).

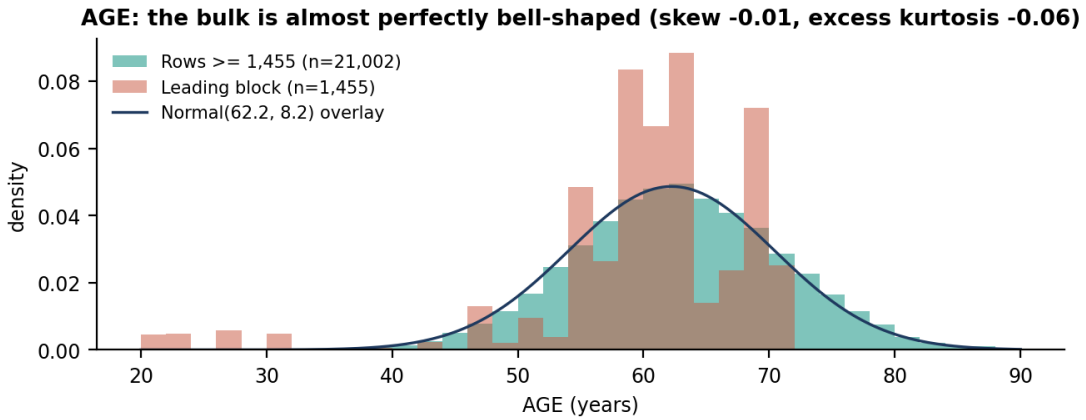


Figure 4. AGE distribution of the leading block versus the rest, with a Normal(62.2, 8.2) overlay.

How to talk about this on stage

Say: 'the structure is consistent with programmatic generation or resampling, and is unlike what we would expect from real self-reported clinical data.' Do **not** say 'the dataset is fake'. We have not established provenance; we have established statistical behaviour.

4.6 Sensitivity analysis: four analysis views

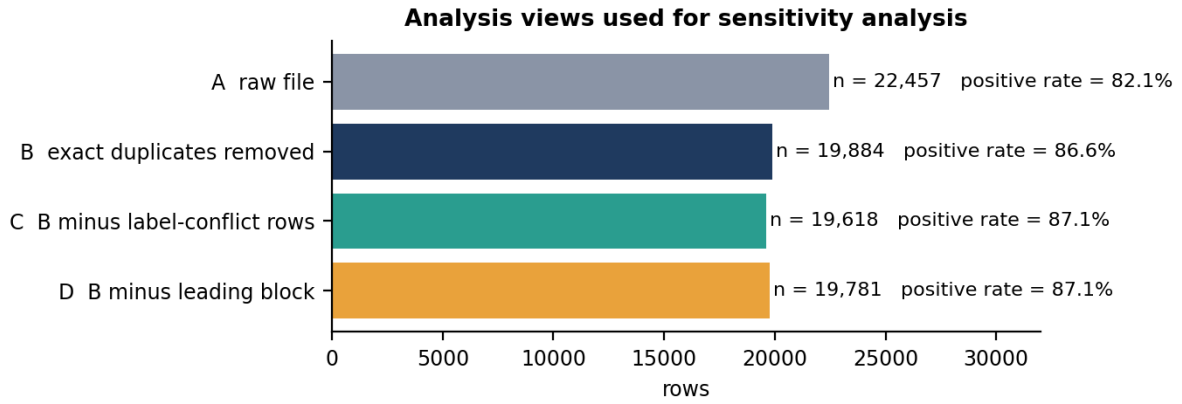


Figure 5. The four analysis views. A = raw. B = exact duplicates removed (keep first). C = B without rows whose feature pattern has conflicting labels. D = B without the leading block.

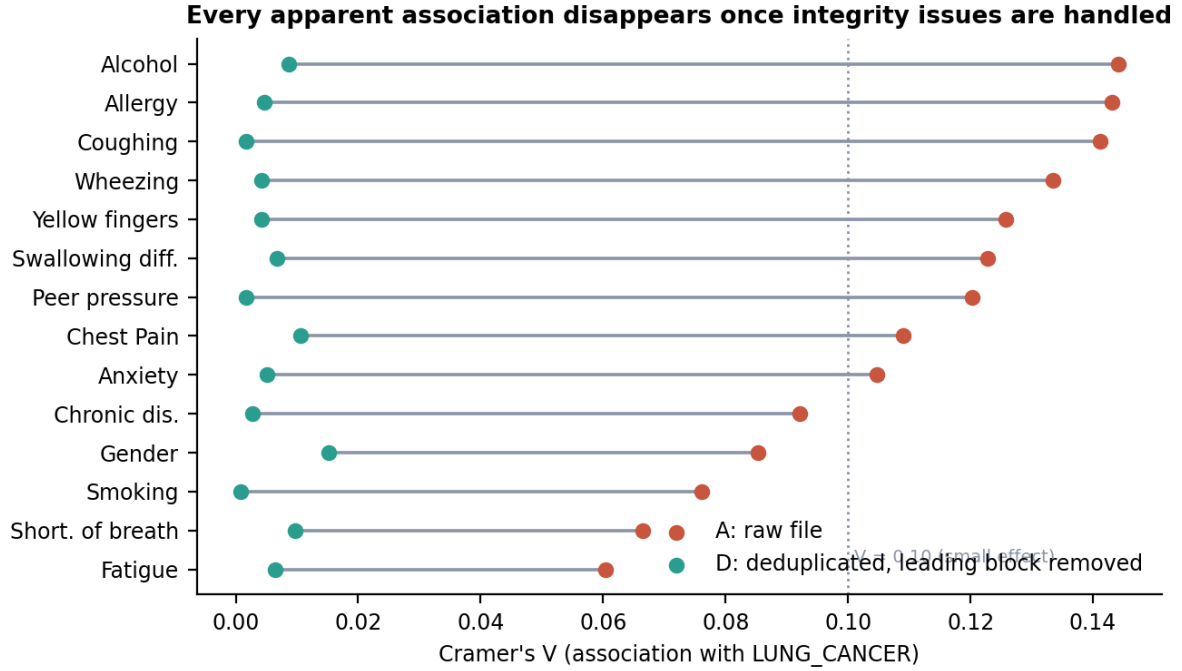


Figure 6. Cramer's V per feature in view A (raw) and view D. Every apparent association shrinks to below 0.016.

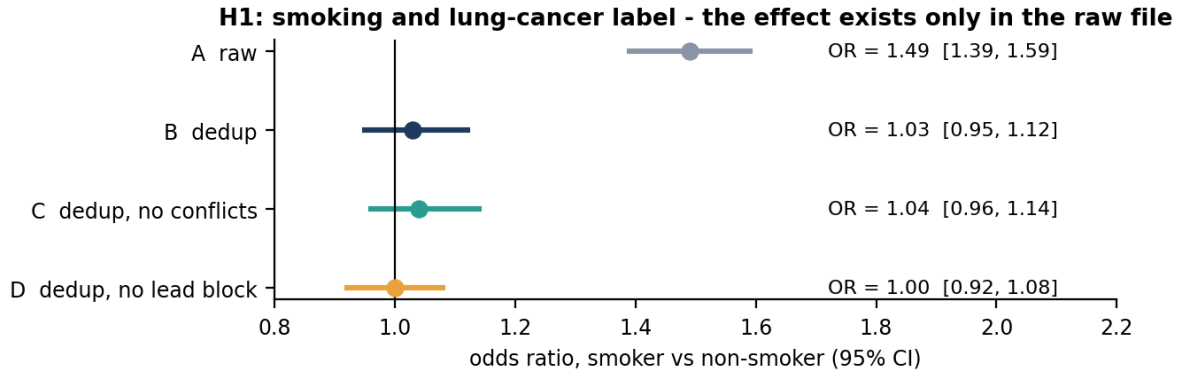


Figure 7. H1 (smoking and the recorded label): odds ratios with 95% Wald CIs per view. The raw-file effect (OR 1.49) disappears (OR 1.00 in D).

View	n	Positive rate	P(pos | smoker)	P(pos | non-smoker)	OR [95% CI]	chi-square p	V
A raw	22,457	82.1%	0.848	0.789	1.49 [1.39, 1.59]	4.5e-30	0.076
B dedup	19,884	86.6%	0.868	0.864	1.03 [0.95, 1.12]	0.52	0.005
C no conflicts	19,618	87.1%	0.873	0.868	1.04 [0.96, 1.14]	0.30	0.007
D no lead block	19,781	87.1%	0.870	0.871	1.00 [0.92, 1.08]	0.91	0.001

- **Family test in view D** (14 chi-square tests, Benjamini-Hochberg): smallest p = 0.033 (GENDER code), q = 0.46; every q is at least 0.46. AGE (Mann-Whitney) p = 0.051, common-language effect size 0.512 (0.5 = no effect).
- **Power**: with 11,080 smokers and 8,701 non-smokers in view D, the minimum risk difference detectable at 80% power and alpha 0.05 is about 1.3 percentage points. The null is therefore informative: any smoking effect larger than about 1.3 points would very likely have been detected. This is a statement about this file, not about real smokers.
- **Joint test** (likelihood ratio, logistic regression with all 15 features vs intercept only; used as inference, not as a predictive model): view B chi-square 26.5 on 15 df, p = 0.033, McFadden pseudo-R2 0.0017; view D chi-square 17.1, p = 0.31, pseudo-R2 0.0011. Even where nominally significant, the explanatory power is negligible.
- In view A, AGE of positives is 62.28 vs 61.13 years (Mann-Whitney p = 6.7e-9); in D the gap is 62.27 vs 61.94 (p = 0.051). The 'under 40' positive rate is 43% in A but 83% in D (n = 57), another artefact.

> **Reading the result correctly**
>
> The absence of an association between smoking and the label is itself evidence of a **construct-validity failure**: in real populations smoking is a strong cause of lung cancer (USPSTF). A dataset in which smokers and non-smokers have the same recorded rate cannot be a faithful record of lung-cancer status. That statement concerns the file, not smoking.

4.7 Internal leakage diagnostic (not part of the submission)

The rubric excludes model training, so the following numbers are an **internal sanity check to understand the risk**, not a result to present as modelling. Five-fold ROC AUC, logistic regression (LR) and random forest (RF), default settings:

Evaluation scheme	LR	RF	Reading
Raw file, shuffled stratified folds	0.671	0.695	Inflated: duplicates and the leading-block shortcut.
Raw file, shuffled, grouped by identical feature pattern	0.672	0.659	Still above chance: the block's distribution shift is learnable even for unseen patterns (source confounding).
Raw file, contiguous unshuffled folds	0.433	0.477	Below chance: row order plus block structure break the folds.
Duplicates removed, shuffled stratified folds	0.499	0.503	Chance level. No learnable signal in the cleaned data.
patient_id only (Mann-Whitney AUC)	0.670	-	The identifier is a label proxy.

4.8 Competing explanations and how to discriminate them

Explanation	Predicts	Test (owner)
H-S: composite file: a small resampled real-survey block plus a generated bulk with random labels	Leading patterns match the public 309-row survey; bulk has independent columns	B-4 lineage check; E independence tests (done)
H-G: wholly generated	No match to any public file; independence everywhere	B-4 lineage check
H-L: real data with a labelling rule unrelated to these features	Correlated symptom structure inside the bulk	Contradicted: bulk is uncorrelated (r max 0.022)
H-E: our audit is wrong	Independent re-implementation gives different counts	C and F each re-implement the headline numbers without looking at audit.py

5. Decision log

Decisions are recorded once, here. A change requires a line in the log (date, who, why). Status: **[DECISION]** adopted unless marked proposed.

ID	Decision	Rationale	Revisit if
D-01	data/raw is read-only (chmod 444), SHA-256 prefix 4ad2333067fcb0a9 is asserted at load time. All changes happen in code, never in the file.	Reproducibility; rubric 'before/after checks'.	Never.
D-02	Integrity-first framing: the project question is whether the file carries reproducible information, not 'how accurate is a model'.	Evidence (Ch. 4); matches the rubric's emphasis on diagnosis and leakage.	Provenance shows the label is clinical and the file is genuine.
D-03	Primary analysis view = B (exact duplicates removed, keep first). C and D are pre-specified sensitivity views. A appears only to quantify the artefact.	Minimal, transparent intervention; conclusions are identical in B, C, D.	Instructor or source requires keeping duplicates.
D-04	patient_id and any row-position information are excluded from every feature set.	AUC 0.67 from the identifier alone.	Never.
D-05	No model is trained in the midterm deliverable; modelling is described as planned work.	Rubric excludes model training.	Instructor extends scope.
D-06	GENDER is reported as GENDER_CODE (0/1) until the source confirms the mapping.	Unverified encoding; avoids a false claim about sex.	B-4 resolves it.
D-07	One repository, one Makefile, seed 42, every figure and number regenerated by code; slides and report quote only results files.	Consistency across code, report, presentation.	Never.
D-08	AGE outliers (221 rows by IQR rule, 123 under 40) are kept and flagged.	Plausible ages, no sentinel values; sensitivity covered by view D which drops the young block.	Source says ages are impossible.
D-09	Vocabulary: 'recorded label', never 'diagnosis'; 'risk indication', never 'detection'.	Construct-validity risk (Ch. 3.2).	Provenance confirms clinical diagnosis.
D-10	Hypotheses H1 to H5 are fixed before running the final tests (Ch. 6.4); alpha 0.05; BH correction across the family; effect sizes and CIs always reported.	Prevents p-hacking; the large n makes everything significant otherwise.	Never.
D-11	Report written in a shared document (Google Docs or Word) and exported to PDF; figures come only from reports/figures. Proposed.	Seven authors need parallel editing.	Instructor mandates LaTeX or Jupyter-only.
D-12	Pinned environment (requirements.txt with exact versions after the first successful run), Python 3.11 or newer.	Reproducibility.	A member cannot install.

6. CRISP-DM walkthrough: build, learn, validate

CRISP-DM (Chapman et al., 2000) describes a data-mining project as six phases (business understanding, data understanding, data preparation, modelling, evaluation, deployment) with loops back to earlier phases. The rubric splits data preparation into three graded parts. We follow the **rubric numbering** and note the CRISP-DM name. **[SOURCED]**

CRISP-DM as applied here: the midterm covers phases 1-3 and previews 4-6

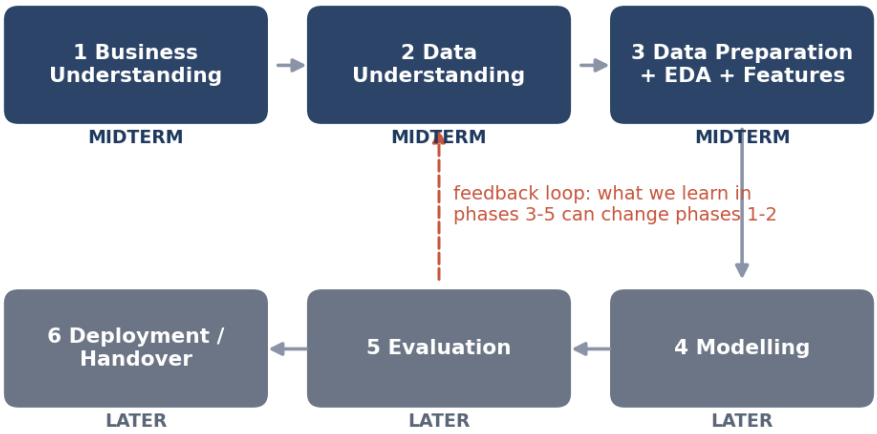


Figure 8. CRISP-DM as applied to this project. The feedback loop is real: the audit in phase 3 changed the research questions of phase 1.

Rubric phase	CRISP-DM phase	Key outputs	Owner(s)	Code / document
1 Business understanding	Business understanding	Problem statement, 5 research questions, success criteria, scope	A	report s.1, slides 2-3
2 Data understanding	Data understanding	Dictionary, provenance, schema tests, first quality notes	B	src/data_io.py, docs/data_dictionary.md
3 Preprocessing	Data preparation	Audit, cleaning policy, views A-D, before/after log, leakage rules	C, D	src/audit.py, src/clean.py
4 EDA	Data understanding (deeper)	Figures, correlations, hypothesis tests H1-H5	E, F	notebooks/03_eda.ipynb, src/stats_utils.py
5 Feature engineering	Data preparation (final dataset)	Feature spec, three feature sets, verification	G	src/features.py, docs/feature_spec.md
6 Conclusions and next steps	Evaluation (of the data, not a model) and planning	Findings, limitations, modelling plan	A, G	report s.6, Ch. 14

Phase 1 - Business understanding (owner A, due Sun 25 Oct noon)

Purpose. Convert a vague topic ('predict cancer') into a precise, answerable, measurable question, and decide what success and failure look like *before* touching the data further. The rubric asks for context, a precise problem statement, focused research questions and explicit success criteria.

Concepts to master

- **Business goal vs data-science goal.** Goal: a trustworthy symptom-based risk indication. Data-science goal: establish whether the file supports it.
- **SMART success criteria**: specific, measurable, achievable, relevant, time-bound. 'Good accuracy' is not measurable; '100% of cleaning steps logged with before/after counts' is.
- **Construct validity**: does the recorded label measure the disease we say it measures?
- **Prior knowledge as a hypothesis source**: smoking and age should matter; if they do not, that is informative.

Problem statement (adopt as written, edit wording only)

Lung cancer is the leading cause of cancer death, and risk indication tools in practice rely on detailed smoking history and age. We study a 22,457-row public-style survey file of self-reported symptoms and habits with a recorded lung-cancer label. Our goal is to determine whether this file contains reproducible information linking those answers to the recorded label, to document and correct data-integrity problems that would invalidate any later model, and to prepare a leakage-safe analysis foundation. This is a research prototype and makes no clinical claim.

Research questions

ID	Question	Answered in	Phase
RQ1	What integrity problems (order artefacts, duplicates, label conflicts, source shift, identifier leakage) does the file contain, and how large are they?	Ch. 4, report s.3	2, 3
RQ2	After integrity treatment, which features, if any, are associated with the recorded label, and with what effect size and uncertainty?	H1-H5, report s.4	4
RQ3	How sensitive are those conclusions to the cleaning policy (views A to D)?	Fig. 5-7, report s.4	3, 4
RQ4	How representative is the sample (age, prevalence, feature structure) of a plausible screening population?	Ch. 3-4, report s.2	2
RQ5	Which engineered features are justified by domain knowledge or evidence, and which are not?	Report s.5	5

Success criteria (measurable; tick them on Tuesday)

ID	Criterion	How verified
S1	Raw file unchanged: SHA-256 prefix 4ad2333067fcb0a9 asserted by tests.	pytest passes on a clean clone
S2	Every transformation is a function with a logged before/after row count.	reports/tables/cleaning_log.csv
S3	All numbers in report and slides regenerate from `make all` in under 5 minutes.	A runs it on a fresh machine Tuesday
S4	H1 to H5 each reported with test, effect size, 95% CI, p and BH q; one primary hypothesis (H1) with permutation check.	Report s.4 table
S5	Sensitivity analysis on at least four cleaning views.	Fig. 5-7
S6	No identifier or label-derived column in any feature set; asserted by tests.	tests/test_pipeline.py
S7	Presentation finishes in at most 19:30 in two timed rehearsals.	A's rehearsal log
S8	Each member answers at least 80% of mock Q and A questions on all phases.	Mock round scorecards
S9	Zero clinical-claim phrases in report and slides (banned list, Ch. 12).	A's text search before freeze

Scope and non-goals

- In scope: audit, cleaning, EDA, statistics, feature engineering, modelling plan.
- Out of scope for the midterm: model training or tuning, clinical interpretation, deployment, any claim of diagnostic value.

Self-check. (1) Why is 'achieve 90% accuracy' a bad success criterion on this file? (2) What would convince you that the label measures disease? (3) Which RQ changed after the audit and why?

Phase 2 - Data understanding (owner B, due Sun 25 Oct evening)

Purpose. Know exactly what each column is, where it came from, and what the first quality problems are, before cleaning anything. Output: loader, schema tests, data dictionary, data card.

Concepts to master

- **Measurement scale**: GENDER and symptoms are nominal/binary, AGE is ratio-scale. A 1/2 coding is a label, not a number: averaging it is allowed only as 'share of 2s'.
- **Data provenance**: who collected it, how, when, under what consent and licence. If unknown, the correct answer is 'unknown' (OPEN), not a guess.
- **Schema validation**: assertions on columns, dtypes, ranges and uniqueness. Cheap, and catches silent file changes.
- **Sentinel values and hidden missingness**: 999, -1, 0 or blank strings can mean 'missing'. Check every column's value set.

Steps

- Create the repository and environment (Ch. 8). Copy the CSV into data/raw, `chmod 444`, record `sha256sum` in data/raw/CHECKSUMS.txt.

- Implement `load_raw()` with fingerprint check and `validate_schema()` (Appendix B). Run it; expect (22457, 17), zero nulls, all int64, symptoms in {1,2}, GENDER and label in {0,1}.
- Write docs/data_dictionary.md: one row per column with type, coding, units, value range, share of YES, and 'unknown' where documentation is missing.
- Write the data card: source URL (OPEN), licence (OPEN), collection method (as described publicly), known limitations, our SHA-256, date of access.
- Task B-4: download the original 309-row Kaggle file and the 3,000-row file; test whether their rows occur in ours (exact match on the 15 features after recoding GENDER M/F to 0/1 both ways); report match counts.
- Initial quality notes (feed to phase 3): 0 nulls, 22,457 unique ids, 2,573 duplicates, 133 conflicting patterns, ordered file.

```
# docs/data_card.md (skeleton)
Name: lung_cancer_combined_22457.csv         Accessed: 2026-10-24     SHA-256 prefix: 4ad2333067fcb0a9
Provided by: <instructor / team member / URL>   Licence: <UNKNOWN - ask>   Redistributable in repo: <yes/no>
Unit of observation: one row per recorded respondent (identity unknown)   Rows: 22,457   Columns: 17
Target: LUNG_CANCER (0/1) - meaning unverified; do not call it diagnosis
Known issues: ordered file; duplicated all-negative leading block; exact duplicates; label-conflicting patterns
Not available: smoking dose/duration/quit time, imaging, biomarkers, dates, geography, ethnicity
```

Self-check. (1) Why do we not treat the 1/2 flags as numbers in a regression without recoding? (2) What does a unique, sequential patient_id tell you about the file's construction? (3) Which two facts from phase 2 already contradict the 'early detection' claim?

Phase 3 - Data preprocessing / preparation (owners C and D, due Mon 26 Oct noon)

Purpose. Diagnose, then treat, every quality problem with a defensible rule, log the effect, and make leakage impossible by construction. This phase carries 12 code points and 5 report points: it is the main place to outscore other teams.

Concepts to master

- **Missingness mechanisms**: MCAR (random), MAR (depends on observed data), MNAR (depends on the missing value itself). Not needed here (no nulls) but you must be able to explain why and what you would do (imputation fitted on training folds only).
- **Duplicates**: exact vs near; pseudo-replication; train/test contamination. Policy must be stated, applied before any split, and tested for sensitivity.
- **Label noise and conflicts**: identical inputs with different labels cap achievable accuracy and may indicate a labelling process unrelated to the inputs.
- **Outliers**: statistical (IQR fences: Q1 - 1.5 IQR, Q3 + 1.5 IQR) vs domain-impossible. Remove only the impossible; flag the merely unusual.
- **Source confounding**: when part of a file comes from a different source, the source can masquerade as a predictor (a form of Simpson's paradox).
- **Leakage** (Ch. 7.6): any information at training time that would not be available, or independent, at deployment or test time.

Treatment table (fill the 'result' column from your own run)

Issue	Diagnostic	Treatment (decision)	Result to log
Missing values	isna().sum(), value sets, sentinel scan	None needed; document that none exist and that no sentinel codes were found	0 nulls
Types and categories	dtype table, value sets	Recode flags 1/2 to 0/1 (2 = YES) in features.py; GENDER to GENDER_CODE (D-06)	all 13 flags in {0,1}
Exact duplicates	duplicated(features + label)	Remove, keep first (view B, D-03)	22,457 to 19,884 (-2,573)
Label conflicts	groupby(features).label.nunique() > 1	Keep in B; remove in sensitivity view C; report both	133 patterns, 315 rows; C = 19,618
Order artefact / source block	index of first positive; per-feature V block vs later negatives	Keep in B; remove in sensitivity view D; shuffle before any split	block 1,455 rows; D = 19,781
Outliers (AGE)	IQR fences, domain range 18-110	Keep and flag (D-08); view D covers the young block	221 flagged
Identifier leakage	AUC of patient_id	Drop patient_id from all features (D-04); test asserts it	AUC 0.67
Integration of sources	lineage check (B-4)	Single file; document upstream combination	match counts

Leakage prevention rules (the rubric names this explicitly)

Rule	Why	How we enforce it
L1 Deduplicate and group before splitting.	Copies in train and test inflate scores.	views built in clean.py; future splits grouped by feature pattern
L2 Shuffle with a fixed seed before splitting; never use contiguous folds on this file.	Order encodes the label.	seed 42; test that fold label rates are similar
L3 Fit every learned transformation (scaler, imputer, encoder, selector) on training data only.	Test statistics must not influence training.	fit_age_scaler(train) pattern in features.py
L4 No identifier, row position or label-derived field as a feature.	Direct proxies of the label.	verify() and pytest assert absence
L5 Feature selection uses training folds only.	Selecting on all data leaks.	midterm uses selection only descriptively; modelling will nest it in CV
L6 Record every count before and after each step.	Auditability.	cleaning_log.csv

Core code (full tested version in Appendix B)

```
def make_views(df):
    fp = leading_block_end(df)                                          # 1455: index of first positive
    is_dup = df.duplicated(subset=FEATURES + [TARGET], keep="first")
    B = df[~is_dup]                                                     # view B: exact duplicates removed
    n_labels = df.groupby(FEATURES)[TARGET].transform("nunique")        # >1 => conflicting labels
    C = B[n_labels[B.index] == 1]                                       # view C: B without conflicts
    D = B[B.index >= fp]                                                # view D: B without the leading block
    return {"A_raw": df, "B_dedup": B, "C_dedup_noconflict": C, "D_dedup_nolead": D}
```

Expected output (make clean): A 22,457 rows 82.09% positive; B 19,884 rows 86.61%; C 19,618 rows 87.11%; D 19,781 rows 87.06%. If yours differ, your duplicate definition differs: use features plus label, keep first.

> **Why remove duplicates before, not after, the train/test split**
>
> If copies of a row land on both sides of the split the model is graded on rows it has memorised. Removing duplicates first (and grouping identical feature patterns in any CV) makes that impossible. Removing them after splitting changes the test set after the fact and still leaves cross-contamination.

Acceptance (C and D). make clean produces the four views; cleaning_log.csv matches the expected counts; pytest passes; a one-slide before/after table exists; each treatment has a one-sentence rationale in the report. **Self-check.** (1) Explain in one sentence why V = 0.54 between the leading block and later negatives matters more than any V with the label. (2) Why keep conflicting rows in the primary view? (3) What would you do if AGE contained 999?

Phase 4 - Exploratory data analysis and statistics (owners E and F, due Mon 26 Oct 18:00)

Purpose. Describe distributions, relationships and structure with purposeful plots, and test pre-registered hypotheses with effect sizes. 13 code points and 6 report points.

Concepts to master (details in Ch. 7)

- Describe before you test: counts, shares, quantiles, skewness, kurtosis per variable and per class.
- Association metrics by scale: chi-square and Cramer's V (binary-binary), Mann-Whitney with common-language effect size (binary-numeric), phi/Pearson (binary-binary correlation), Spearman (ordinal).
- p-values answer 'is the data surprising if there is no effect?', not 'is the effect important?'. With n of 20,000 trivial effects are significant; hence effect sizes, CIs, and the minimum detectable effect.
- Multiple comparisons: 14 tests at 5% give about 0.7 false positives by chance; correct with Benjamini-Hochberg.

Figure plan (7 core figures; every figure answers a written question)

Fig	Question	Content	Owner
E1	Is the file ordered?	Rolling positive rate and feature shares vs row (Fig. 1)	C
E2	How much data survives each policy?	Four views bar chart (Fig. 5)	D
E3	Is the leading block a different source?	Share of YES per feature: block vs later negatives vs later positives (Fig. 2)	E
E4	What is the shape of AGE?	Histogram by block with Normal overlay; skewness and kurtosis (Fig. 4)	E
E5	Are features related to each other?	Correlation heatmap with clipped scale; PCA scree in text (Fig. 3)	E
E6	Does any feature relate to the label, and does it depend on cleaning?	Cramer's V dumbbell A vs D (Fig. 6)	F
E7	Does smoking relate to the label?	Forest plot of OR across views (Fig. 7)	F

Hypotheses (fixed now; D-10)

ID	Null hypothesis and rationale	Test	Effect size	Assumptions to check
H1 (primary)	P(label = 1 | smoker) = P(label = 1 | non-smoker). Rationale: smoking is the dominant known risk factor.	Chi-square (2x2) with permutation check, two-sided	Odds ratio with 95% CI, risk difference	Expected cell counts at least 5; independence of rows (hence deduplicate)
H2	AGE distribution is the same in positives and negatives. Rationale: age raises risk.	Mann-Whitney U (AGE only mildly non-normal but ranks are safer)	Common-language effect size	Independent samples; similar shape for a location interpretation
H3	Respiratory symptom count is unrelated to the label.	Spearman rank correlation or Cochran-Armitage trend	rho; positive rate by level	Ordinal 0-4 variable
H4	None of the 14 binary features is associated with the label (family test).	14 chi-square tests with BH-FDR	Cramer's V per feature	BH assumes independence or positive dependence: reasonable here
H5 (integrity)	Leading-block negatives and later negatives have the same feature distributions.	14 chi-square tests block vs later negatives	Cramer's V	Group sizes 1,455 and 2,568

Method notes

- **Primary view = B.** Run every test on A, B, C, D and tabulate: the stability of the conclusion is itself a result.
- **Report effect sizes with intervals.** Rule of thumb for Cramer's V on 2x2 tables: 0.1 small, 0.3 medium, 0.5 large (Cohen). Our later-row V values are below 0.02: negligible.
- **State power.** View D has 11,080 smokers and 8,701 non-smokers: minimum detectable risk difference about 1.3 percentage points.
- **Never say 'proves' or 'causes'.** Association in an observational file is not causation, and here not even a reliable association.

```
from stats_utils import odds_ratio_ci, permutation_p, feature_family
y, x = B[TARGET], (B["SMOKING"] == 2).astype(int)
print(odds_ratio_ci(x, y))            # (1.027, 0.946, 1.115) on view B
print(permutation_p(x, y, 5000))      # about 0.51
print(feature_family(B.assign(**{c: (B[c] == 2).astype(int) for c in BINARY_12}), BINARY_12))
```

Acceptance (E and F). Seven figures with captions and one takeaway sentence each; hypothesis table filled for all four views; every test reports statistic, p, q, effect size and CI; a half-page 'what the statistics do and do not show'. **Self-check.** (1) Why is a smoker/non-smoker difference of 0.3 points not evidence of a real effect even if p were 0.01? (2) What does CLES = 0.512 mean? (3) Why use BH rather than Bonferroni here?

Phase 5 - Feature engineering (owner G, due Tue 27 Oct 12:00)

Purpose. Build the final feature set from cleaned data, justify each feature, verify it, and prevent leakage. 7 code points, 3 report points.

Concepts to master

- **Encoding**: binary flags to 0/1; nominal multi-level variables to one-hot; ordinal to integer ranks only when order is meaningful.
- **Scaling**: z-score (mean 0, sd 1) matters for distance, kernel and gradient-based models, not for trees. The mean and sd are **learned parameters** and must come from training data only (L3).
- **Transformations**: log or Box-Cox for strong skew; AGE skewness is -0.006, so none is needed (state this explicitly).
- **Construction**: counts, groups, bins, interactions, driven by domain knowledge rather than by trying everything.
- **Selection families**: filter (univariate statistics), wrapper (search with a model), embedded (L1, trees). Always nested inside cross-validation when a model is involved.
- **Extraction**: PCA or MCA compress correlated features. Here the PCA spectrum is flat (first component 7.0% vs 6.7% expected for no structure), so extraction cannot help.

Feature specification (docs/feature_spec.md)

Feature	Definition	Justification	Type
13 flags recoded	(value == 2) to 0/1	Standard binary encoding; keeps the meaning 'YES = 1'	raw
GENDER_CODE	GENDER renamed, unchanged	Mapping unverified (D-06); never reported as sex	raw
AGE, AGE_Z	Raw age; z-score using a scaler fitted on training data	Needed by scale-sensitive models; no skew correction needed	raw / scaled
AGE_BAND	<50, 50-59, 60-69, 70-79, 80+	For plots and subgroup reporting; follows decade bands and the USPSTF 50-80 range	derived (reporting)
IN_SCREEN_AGE	50 <= AGE <= 80	Domain: USPSTF 2021 age range	derived (domain)
RESP_SYMPTOM_COUNT	COUGHING + WHEEZING + SHORTNESS_OF_BREATH + CHEST_PAIN (0-4)	Domain grouping of respiratory symptoms; reduces 4 sparse flags to one ordinal burden score	derived (domain)
SYMPTOM_COUNT_ALL	Respiratory four + FATIGUE + SWALLOWING_DIFFICULTY (0-6)	Overall symptom burden; correlation 0.83 with the respiratory count, so do not use both in a linear model	derived (domain)
SMOKER_IN_SCREEN_AGE	SMOKING AND IN_SCREEN_AGE	Proxy for the age and smoking part of USPSTF eligibility; lacks pack-years and quit time, so it is a proxy only	derived (interaction)
Excluded	patient_id; any row-position feature	Leaks the label (AUC 0.67)	-

Feature set	Columns	Purpose
F0_raw15	15 raw features (flags 0/1, GENDER_CODE, AGE)	Baseline: no engineering
F1_raw_plus_engineered	19: F0 with AGE_Z plus IN_SCREEN_AGE, RESP_SYMPTOM_COUNT, SYMPTOM_COUNT_ALL, SMOKER_IN_SCREEN_AGE	Tests whether engineering adds information
F2_prior_driven	5: AGE_Z, SMOKING, IN_SCREEN_AGE, RESP_SYMPTOM_COUNT, SMOKER_IN_SCREEN_AGE	Minimal, domain-driven set; protects against fishing

Selection protocol and what the data say

- Rule **[DECISION]**: a feature enters the modelling set if it is **domain-justified** (F2) or passes a Benjamini-Hochberg filter inside training folds (future). All 15 raw features remain candidates (F0, F1) for comparison.
- Univariate screening on view B (16 features incl. engineered binaries, chi-square with BH): smallest p = 0.029 (SHORTNESS_OF_BREATH, V = 0.016, q = 0.42); no feature has q below 0.05. On view D the symptom counts have Spearman rho of -0.002 and -0.005 (p = 0.77 and 0.45).
- Variance filter: every binary flag has share between 47% and 67%, so none is near-constant. Redundancy: max |r| 0.022 among raw features and VIF below 1.03, so no raw feature is dropped for redundancy. Engineered counts are correlated with their components by construction (0.83 between the two counts), which is why feature sets, not a pooled list, are compared.
- **Honest conclusion for the report**: feature selection by evidence cannot discriminate between features in this file because none carries detectable information; the final set is therefore chosen by domain priors (F2) with F0 and F1 retained as comparison sets for future modelling.

```
fe = add_features(B, fit_age_scaler(B.AGE))   # B = primary view; scaler fitted here only for DESCRIPTIVE checks
verify(fe)                                    # asserts: no id, ranges, 0/1 flags, set columns exist
assert abs(fe["AGE_Z"].mean()) < 1e-9 and abs(fe["AGE_Z"].std(ddof=0) - 1) < 1e-9
print({k: len(v) for k, v in FEATURE_SETS.items()})   # {'F0_raw15': 15, 'F1_raw_plus_engineered': 19, 'F2_prior_driven': 5}
print(fe.RESP_SYMPTOM_COUNT.value_counts().sort_index().to_dict())   # {0: 663, 1: 3416, 2: 7110, 3: 6426, 4: 2269}
```

Acceptance (G). feature_spec.md complete; verification asserts pass; one slide with the table above; a paragraph that states honestly that selection by evidence is inconclusive and why. **Self-check.** (1) Why must AGE_Z's mean and sd come from the training fold in modelling? (2) Why is SMOKER_IN_SCREEN_AGE a 'proxy' and not 'eligibility'? (3) When would PCA be useful and why is it not here?

Phase 6 - Conclusions and next steps (owners A and G, due Tue 27 Oct 12:00)

Purpose. Summarise evidence, revisit each research question, state limitations precisely and propose the modelling plan without implementing it.

RQ	Draft answer (confirm with your own run before use)
RQ1 integrity	Three artefacts: an ordered, duplicated all-negative block of 1,455 rows (103 patterns), 2,573 exact duplicates and 133 conflicting patterns, and an identifier that proxies the label (AUC 0.67). Later rows behave like independent draws.
RQ2 association	After treatment no feature is associated with the label: largest V = 0.015; smoking OR = 1.00 (0.92 to 1.08); no BH q below 0.46.
RQ3 sensitivity	The apparent associations of the raw file exist only in view A; B, C and D agree. Conclusion is robust to the cleaning policy among B, C, D.
RQ4 representativeness	Not representative: prevalence 82% to 87%, no dose variables, structure unlike real survey data.
RQ5 features	Domain-driven features are defined and verified, but no evidence-based selection is possible; modelling will compare F0, F1, F2 with nested validation.

Limitations to state (specific, not generic)

- Provenance and licence unknown; label meaning unverified; GENDER mapping unverified.
- Statistical behaviour consistent with generation or resampling; a lineage check against the public survey was not possible offline (B-4 pending).
- No smoking dose, duration or quit time, so USPSTF eligibility cannot be reproduced.
- Findings concern this file only; they say nothing about whether smoking or symptoms predict lung cancer in people.
- Single file: no external validation is possible.

7. Concept primer: the statistics and leakage you must master

Read this chapter with the code open. Each concept ends with the exact place in our project where you will use it. If you can explain a concept to a teammate without notes, you are ready for Q and A.

7.1 Hypothesis testing in six sentences

- State a **null hypothesis H0** (no association) and an alternative; choose alpha (0.05) *before* looking.
- Compute a **test statistic** that measures how far the data are from H0 (chi-square, U, rho).
- The **p-value** is the probability, if H0 were true, of a statistic at least as extreme as observed. It is not the probability that H0 is true and not the size of an effect.
- **Type I error** = rejecting a true H0 (rate alpha); **type II error** = missing a real effect (rate beta); **power** = 1 - beta.
- **Effect size** says how big; **confidence interval** says how precisely we know it. Report both next to every p-value.
- **Statistical significance is not practical significance.** With n of about 20,000 an association of V = 0.02 can reach p below 0.05 and still explain 0.04% of variance.

7.2 The tests we use

Test / measure	Use when	Formula idea	Interpret	Our use
Chi-square test of independence	Two categorical variables, expected cell counts of at least 5	sum over cells of (O - E)^2 / E	Large chi-square means counts deviate from independence	H1, H4, H5
Cramer's V	Effect size for chi-square	sqrt(chi2 / (n * min(r-1, c-1)))	0.1 small, 0.3 medium, 0.5 large (2x2)	All associations
Odds ratio, Wald CI	2x2 table	OR = ad / bc; CI = exp(ln OR +/- 1.96 * sqrt(1/a+1/b+1/c+1/d))	OR = 1 means no association; CI crossing 1 means not distinguishable from none	H1
Mann-Whitney U	Compare a numeric variable between two groups without assuming normality	rank-sum based	CLES = U / (n1 n0): P(random positive older than random negative); 0.5 = no effect	H2 (AGE)
Spearman rho	Monotone relation, ordinal variable	Pearson on ranks	-1 to 1; near 0 means no monotone trend	H3
Permutation test	Distribution-free check of a difference	shuffle labels, recompute statistic, count extremes	Fraction of shuffles as extreme as observed	H1 check
Benjamini-Hochberg	Many tests at once	sort p; q(i) = min over j >= i of p(j) * m / j	q below 0.05 controls the expected share of false discoveries at 5%	H4
Likelihood-ratio test	Do all features jointly add to an intercept-only logistic model?	2 * (loglik full - loglik null) ~ chi-square(df)	Inference, not prediction	Joint test in Ch. 4

7.3 Power and the minimum detectable effect

For two proportions near p with group sizes n1 and n0, the smallest difference detectable with 80% power at alpha 0.05 is approximately 2.8 * sqrt(p(1-p)(1/n1 + 1/n0)) (2.8 is 1.96 + 0.84). In view D with p = 0.87, n1 = 11,080, n0 = 8,701 this is about 0.013. Use it to turn 'not significant' into an informative statement: *effects larger than about 1.3 points are very unlikely in this file*.

7.4 Multiple comparisons

Test 14 independent true nulls at 5% and the chance of at least one false positive is $1 - 0.95^{14} = 51\%$. **Bonferroni** divides alpha by the number of tests (very conservative). **Benjamini-Hochberg** controls the false discovery rate and is the usual choice for screening many features. In view D the smallest raw p is 0.033; after BH it is 0.46.

7.5 Correlation, association, causation, confounding

- For two 0/1 variables Pearson correlation equals the **phi coefficient**, and |phi| equals Cramer's V on a 2x2 table: our heatmap and V values are consistent.
- Correlation does not imply causation. Even a genuine smoking-label association would not show that smoking causes the label in this file.
- **Confounding by source**: if a block of rows has both a different feature distribution and a different label rate, the pooled data show an association that vanishes inside each block. That is exactly what removing the leading block does (OR 1.49 to 1.00). Always ask: what else differs between the groups I am comparing?
- **Selection and sampling bias**: our file's prevalence (82 to 87%) cannot be generalised to any screening population.

7.6 Data leakage: eight ways to fool yourself

Kapoor and Narayanan (Patterns, 2023) surveyed 17 fields and found leakage affecting 294 papers, then proposed a taxonomy of eight leakage types and 'model info sheets' to detect them. [SOURCED] The list below follows their taxonomy as we understand it and maps each type to a control in our project.

Leakage type	Meaning	In our project	Control
No train/test split	Evaluating on training data	Not applicable in midterm	Future: split before anything is learned
Pre-processing on train and test together	Scaler, imputer or encoder fitted on all rows	AGE_Z scaler	fit_age_scaler(train) only (L3)
Feature selection on train and test together	Choosing features using test labels	Univariate screening	Descriptive only now; nested in CV later (L5)
Duplicate data points	Same row on both sides	2,573 exact copies	Deduplicate; group identical patterns (L1)
Illegitimate features	Features that proxy the label or are unavailable at prediction time	patient_id (AUC 0.67), row order	Excluded and tested (L4)
Temporal leakage	Future information used	No dates in file; cannot assess	Document as limitation
Non-independence of train and test	Same person/site in both	Unknown respondent identity	Document; group by pattern as a proxy
Sampling bias	Sample differs from target population	82-87% prevalence; source block	State; no prevalence-dependent claims

7.7 Metrics you will need later (preview)

- On imbalanced data, **accuracy** is dominated by the majority class: always-positive scores 82% (raw) or 87% (cleaned). Use ROC-AUC with a CI, **PR-AUC** (whose chance level equals prevalence), **balanced accuracy**, Matthews correlation, sensitivity and specificity at a stated threshold.
- **Calibration** (do predicted 20% risks occur 20% of the time?) via calibration curve, Brier score and calibration slope/intercept; **decision-curve analysis** for net benefit at clinically relevant thresholds.
- **Uncertainty**: bootstrap CIs, repeated stratified grouped CV; **robustness**: label-noise injection, feature-drop tests, shifts between views A to D; **subgroup**: GENDER_CODE and age bands, with sample sizes.
- **Reporting**: TRIPOD+AI 27-item checklist (Collins et al., BMJ 2024) and a leakage info sheet.

Self-check. (1) State in words what p = 0.51 for the smoking permutation test means. (2) Why does BH give q = 0.46 when raw p = 0.033? (3) Give an example of confounding by source from our own data. (4) Why is PR-AUC 'chance level = prevalence'?

8. Reproducibility and engineering standards

Code quality carries only 3 points on its own, but reproducibility underpins the other 37 code points and the credibility of every number. The reference implementation described here was built and tested for this handbook: six pytest tests pass, and `make all` runs in under two seconds. **[VERIFIED]**

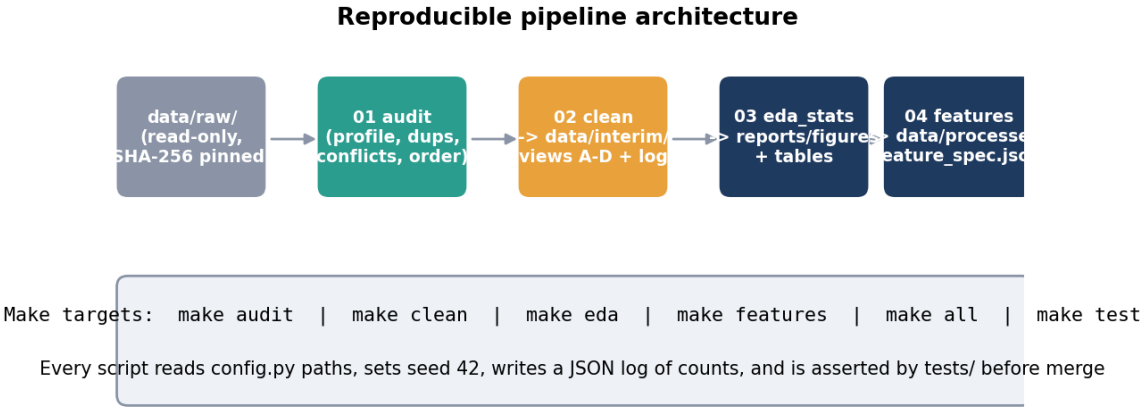


Figure 9. Pipeline architecture: raw data flows one way into interim views, figures and feature specifications. Every arrow is a script.

8.1 Repository layout

```
lung-cancer-ds/
  README.md              # what it is, how to run (5 commands), data provenance, licence note
  Makefile  requirements.txt  .gitignore  .pre-commit-config.yaml (optional)
  data/raw/              # read-only, CHECKSUMS.txt, never edited       <- do not commit if licence forbids; keep checksum + source
  data/interim/          # views A-D (generated)      data/processed/  # feature tables (generated)
  src/                   # config.py data_io.py audit.py clean.py stats_utils.py features.py
  tests/                 # test_pipeline.py (counts, leakage, schema)
  notebooks/             # 01_data_understanding  02_cleaning  03_eda  04_features  (thin: call src/, show outputs)
  reports/figures/ tables/   # generated only; the report and slides use these files
  docs/                  # data_dictionary.md data_card.md feature_spec.md decision_log.md  results.json
  slides/  report/       # source of presentation and report
```

8.2 Environment

```
python -m venv .venv && source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install pandas numpy scipy statsmodels matplotlib seaborn scikit-learn pytest jupyter
pip freeze > requirements.txt                              # pin exact versions after the first successful run
# reference run used: Python 3.12.3, pandas 3.0.2, numpy 2.4.4, scipy 1.17.1, statsmodels 0.15.0, scikit-learn 1.8.0, matplotlib 3.10.8
```

8.3 Rules

- **No absolute paths.** All paths come from src/config.py via pathlib. **Seeds**: 42 everywhere.
- **Notebooks are thin.** They import from src/, display outputs and commentary. Run 'Restart and run all' before every commit; strip outputs with nbstripout or commit with outputs only on main.
- **Generated artefacts are not edited by hand.** Figures and tables come from code; the report and slides copy them verbatim.
- **One results file.** docs/results.json holds every number quoted in report and slides (counts, V values, ORs). G owns it; nobody types a statistic from memory.
- **Tests are the contract.** The test file asserts the expected counts (22,457 / 19,884 / 19,618 / 19,781), the checksum, absence of the identifier, and feature ranges. A failing test blocks the merge.
- **Clean-clone test.** On Tuesday A clones the repository on a fresh folder, installs requirements, runs `make all` and times it.

8.4 Git conventions

Item	Convention
Branches	main is protected. Work on feat/<initial>-<topic>, fix/<topic>, docs/<topic>. Delete after merge.
Commits	Imperative, scoped: 'clean: add label-conflict view C', 'eda: add Cramer V dumbbell'. One logical change per commit.
Pull requests	Small (under about 300 lines), linked to a task ID (e.g. T-D1), template below, reviewed by the assigned partner then merged by A (or by the reviewer after 21:00 integration).
Review pairs	B with C, D with E, F with G; A reviews anything touching results.json or the decision log. Review within 4 hours during the sprint.
Data and secrets	Never commit credentials. Commit data only if the licence allows; otherwise commit CHECKSUMS.txt and the source link.
Releases	Tag v0.1-midterm at the freeze; the submission is built from that tag only.

```
# .github/pull_request_template.md
Task ID:            Owner:            Reviewer:
What changed (1-3 lines):
Evidence: command run + the key numbers it printed
[ ] make all and pytest pass locally      [ ] no absolute paths      [ ] figures/tables regenerated, not hand-edited
[ ] numbers added to docs/results.json    [ ] no clinical-claim wording      [ ] notebook restarted and run top to bottom
```

8.5 Definition of done (for any task)

- Code runs from a clean clone through make; tests cover the new logic.
- Output files exist under reports/ or docs/ and are referenced by name in the report outline.
- A reviewer other than the author has re-run it and agrees with the printed numbers.
- The author can explain the code line by line and the statistics in plain language.

9. Team operating system

Seven people, four days, three graded artefacts. The system below removes ambiguity about who does what, when work is integrated, and how problems are escalated. Names are placeholders: assign people to letters A to G in the first 15 minutes (**[DECISION]**).

9.1 Roles

Letter	Role	Accountable for	Cross-trained backup
A	Team lead, phase 1 and 6, integration, presentation director	Decision log, open questions to the instructor, merge gate on main, rehearsals, final submission	G
B	Data understanding and provenance	Loader, schema tests, dictionary, data card, lineage check (B-4)	C
C	Integrity audit	Order, duplicates, conflicts, source shift, identifier leakage; independent re-implementation of headline numbers	B
D	Cleaning pipeline and tests	make_views, cleaning log, pytest, Makefile, clean-clone test	E
E	Exploratory analysis and figures	Descriptives, correlation, figures E3 to E5, figure style	D
F	Statistical inference	H1 to H5, effect sizes, BH, power, permutation checks, figures E6 and E7	G
G	Feature engineering and report editor	feature_spec, feature sets and verification, results.json, report assembly and consistency	A and F

9.2 Rhythm for the four days

When	What	Who	Output
Sat 24, 15 min kickoff	Read Ch. 1, 2 and 4; assign letters; confirm decisions D-01 to D-12; set up repo	all	Repo with raw data, Makefile, branch protection
Daily 20:30, 15 min standup (video or chat)	Each person: done / next / blocked. Blockers over 2 hours escalate to A	all	Standup notes in docs/standup.md
Daily 21:00 integration checkpoint	Merge approved PRs, run make all and pytest on main, update results.json	D (run), A (merge)	Green main or an issue per failure
Sun 25 18:00 interim review	Phases 1 and 2 complete; audit numbers reproduced independently by C and F	A, B, C, F	Go or no-go on the integrity-first story
Mon 26 18:00 content freeze	All figures, tables and tests final; report draft complete; slides v1	all	Tag v0.9
Mon 26 evening: rehearsal 1	Full run with a timer; no slides editing during the run	all	Timing log, issue list
Tue 27 morning: Q and A round 1	35-question bank, cross-assigned; scorecards	all	Scorecards, weakest topics
Tue 27 afternoon: rehearsal 2 and clean-clone test	Target 19:30; a fresh machine runs make all	A, D	Rehearsal log, clean-clone log
Tue 27 23:59 freeze	Tag v0.1-midterm; export report PDF and slides PDF; only typo fixes after	A	Submission folder
Wed 28 morning	Final run-through of slides only (20 min), logistics check, backup laptop	all	Present

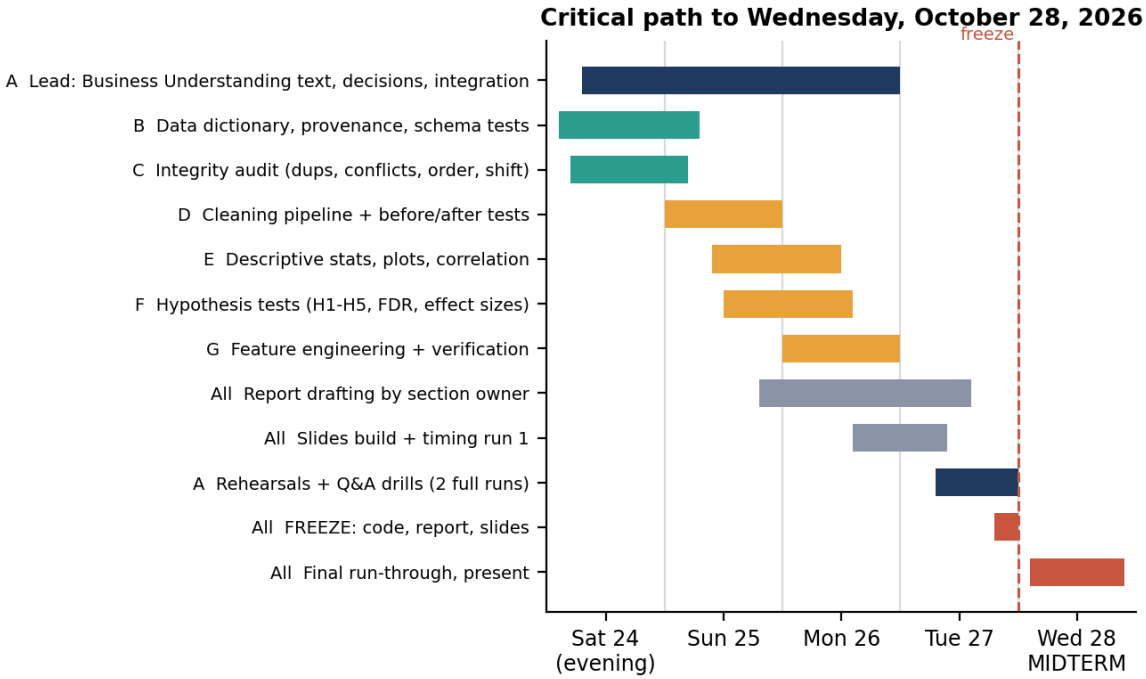


Figure 10. Critical path. The only hard dependencies: audit (C) before cleaning (D); cleaning before EDA, tests and features; everything before the freeze.

9.3 Communication and decision rules

- One channel for decisions (decision log), one for chat. 'Decided in chat' does not exist until it is in docs/decision_log.md.
- **Decision rights.** Technical choices inside a task: owner. Anything that changes a number in results.json, a figure, or a rubric item: owner plus reviewer. Anything that changes the story or a decision ID: A after a 10-minute discussion; if unresolved, A decides and logs the dissent.
- **Work-in-progress limit.** At most two open branches per person; finish before starting.
- **Escalation.** Blocked over 2 hours: say so in chat with what you tried. Slipping a deadline by over 3 hours: tell A, who reassigns or reduces scope using the ladder in 11.2.
- **Respect and load.** Review feedback is about code and claims, never people. Everyone presents and everyone answers questions; nobody is 'just the coder'.

10. Task cards with acceptance criteria

Each task has an ID, inputs, outputs, an acceptance criterion a reviewer can check in five minutes, a due time and a reviewer. Times are Tunis time. If a due time is impossible, tell A before it passes.

ID	Who	Task	Depends on	Output / acceptance criterion	Due	Rev.
T-A1	A	Create repository, branch protection, README, decision log; assign letters; email instructor with OPEN questions Q1 to Q9 (App. E)	-	Repo exists; D-01 to D-12 committed; email sent	Sat 24 20:00	G
T-A2	A	Write Business Understanding: problem statement, RQ1 to RQ5, S1 to S9, scope (Ch. 6)	T-C1 headline numbers	One page in report draft; RQs match Ch. 6	Sun 25 12:00	G
T-A3	A	Master slide template, speaker order, timing sheet; rehearsal scripts	T-A2	Slides skeleton with 14 slides; timing sheet	Mon 26 12:00	F
T-A4	A	Run two timed rehearsals and two Q and A rounds; keep logs; clean-clone test with D	all	Rehearsal time at most 19:30; scorecards filed	Tue 27 18:00	-
T-B1	B	Environment, loader with checksum, schema validation, tests for them	T-A1	load_raw() passes; pytest green on schema tests; shape (22457, 17)	Sat 24 23:00	C
T-B2	B	docs/data_dictionary.md (15 features + id + target), data_card.md with every OPEN marked	T-B1	Every column documented; unknowns labelled 'unknown'	Sun 25 14:00	C
T-B3	B	Data Understanding report section (500 to 700 words) and slides 4 to 5	T-B2	Contains dimensions, types, units, key variables, first quality notes	Sun 25 20:00	A
T-B4	B	Lineage check: download original 309-row and 3,000-row Kaggle files (login needed), test row matches against our file with GENDER mapping tried both ways	T-B1	Table of match counts; one paragraph of interpretation	Mon 26 12:00	C
T-C1	C	Audit: first-positive index, lead block patterns, duplicates, conflicts, identifier AUC, source shift; independent code (not copied from audit.py) matching the handbook numbers	T-B1	reports/tables/audit_summary.json equals Ch. 4 numbers; discrepancies reported	Sun 25 12:00	F
T-C2	C	Figures E1 (order) and E3 (prevalence by group) with captions and takeaway lines	T-C1	Two PNGs at 200 dpi, readable at slide size	Sun 25 20:00	E
T-C3	C	Preprocessing report section: findings, rationale, effect (500 to 800 words); slides 6 to 7	T-C1, T-D1	Each issue has diagnostic, treatment, effect, rationale	Mon 26 14:00	D
T-D1	D	make_views with log; cleaning_log.csv; view files in data/interim	T-C1	Counts 22,457 / 19,884 / 19,618 / 19,781; log exists	Sun 25 16:00	C
T-D2	D	pytest suite and Makefile; run make all on a clean clone	T-D1	6 or more tests green; make all under 5 minutes	Mon 26 12:00	B
T-D3	D	Figure E2 (views); before/after table; slide 8 and report cleaning-decisions table	T-D1	Table reproduces counts; each view labelled A to D	Mon 26 12:00	E
T-E1	E	Descriptive tables (per variable and per class) and figures E4, E5 with PCA note	T-D1	Tables as CSV; figures with captions	Mon 26 12:00	D
T-E2	E	EDA report section (600 to 900 words) and slides 9 to 10	T-E1, T-F1	Analytical discussion, not raw output	Mon 26 18:00	F
T-F1	F	H1 to H5 on views A to D: tables with statistic, p, q, effect size, CI; permutation check for H1; power	T-D1	reports/tables/hypotheses.csv; numbers match Ch. 4 within rounding	Mon 26 12:00	C

ID	Who	Task	Depends on	Output / acceptance criterion	Due	Rev.
T-F2	F	Figures E6, E7; statistics report section and slide 11 (including 'what significance does not mean')	T-F1	Forest plot and dumbbell with CIs	Mon 26 18:00	E
T-F3	F	Independent re-implementation of H1 on view B (statsmodels or scipy only)	T-D1	Same OR and CI to 2 decimals	Sun 25 20:00	C
T-G1	G	features.py, feature_spec.md, three feature sets, verification, tests	T-D1	verify() passes; sets of 15, 19, 5 columns	Mon 26 12:00	F
T-G2	G	docs/results.json generator (all quoted numbers) and consistency check against slides and report	T-F1, T-D1	Script prints mismatches; zero at freeze	Mon 26 18:00	A
T-G3	G	Report assembly: template, figure numbering, references, TRIPOD-style limitations paragraph; PDF export	all sections	10 to 14 pages, six sections, captions, references	Tue 27 14:00	A
T-G4	G	Feature engineering report section (400 to 600 words) and slide 12	T-G1	Spec table, selection protocol, honest conclusion	Mon 26 20:00	F

10.1 Cross-training pairs

The rubric awards 8 points for Q and A and requires all members to be ready for technical questions. Each person must be able to answer for their partner's phase: **B-C** (data and audit), **D-E** (cleaning and EDA), **F-G** (statistics and features); A covers everything. Mechanism: on Monday evening each person teaches their partner their code in 15 minutes using only the notebook; on Tuesday the partner is asked the questions first in the mock round.

11. Risks, contingencies and the scope ladder

11.1 Risk register

ID	Risk	L	I	Mitigation	Trigger and fallback	Owner
R1	Schedule slip: four days only	H	H	Task cards, daily integration, scope ladder (11.2)	Any task over 3 hours late: drop 'could' items	A
R2	Numbers differ between members or between code and slides	M	H	results.json, independent re-implementation, consistency script	Any mismatch blocks the merge	G
R3	Instructor challenges the integrity-first story	M	H	Show views A to D side by side; stress that B, C, D agree; emphasise method over verdict	Fallback: present A and B as 'baseline' and 'cleaned' and keep sensitivity as a backup slide	A
R4	Members unable to answer questions	H	H	Cross-training, 35-question bank, two mock rounds	Weakest topic gets a 15-minute teach-back	A
R5	Overclaiming (clinical or causal)	M	H	Banned phrases list, footer disclaimer, review pass on every slide and paragraph	Rewrite with 'recorded label', 'association in this file'	A
R6	Environment mismatch on the demo or grading machine	M	M	requirements.txt pinned, clean-clone test, PDF backups of notebooks and slides	Run from exported HTML or PDF	D
R7	Merge conflicts in notebooks	M	M	Thin notebooks, one owner per notebook, nbstripout	Re-export from src/	D
R8	A member is absent or ill	M	H	Cross-training backups (9.1)	Backup presents that section; reduce speaker count to six	A
R9	Data licence forbids redistribution	M	L	Commit checksum and source, not data, if required	Provide download instructions	B
R10	Presentation overruns 20 minutes	M	M	Timing sheet per slide, two timed runs, cut list	Cut slide 10 content then slide 8 detail; never cut limitations	A
R11	Provenance check shows our audit mislabels the structure	L	M	Wording discipline ('consistent with', not 'is')	Update Ch. 4 interpretation; numbers unchanged	C
R12	A hypothesis is added after seeing the data	M	M	D-10 fixes H1 to H5 now; anything else labelled exploratory	Move to 'exploratory' appendix	F

L = likelihood, I = impact (H high, M medium, L low).

11.2 Scope ladder (what to cut first)

Level	Items	Rule
MUST (never cut)	Loader and checksum; audit numbers; views A to D; cleaning log; H1 with effect size and CI; family test with BH; five core figures; feature spec; report six sections; 14-slide deck; limitations; Q and A prep	If any is at risk on Monday morning, A reassigns people to it
SHOULD	Permutation check; power statement; independent re-implementation; lineage check (B-4); PCA note; AGE_BAND subgroup plot; TRIPOD-style limitations paragraph	Cut in this order if behind: PCA note, AGE_BAND plot, lineage
COULD	Mutual information screening; Cochran-Armitage trend test; extra sensitivity (leave-block-out); poster-style summary figure; data card as separate PDF	Only after MUST and SHOULD are green
WON'T (midterm)	Any model training; hyperparameter tuning; dashboards; deployment; external datasets in the deliverable	Mention only as planned work

12. Written report blueprint (25 points)

Target: 10 to 14 pages PDF ([ASSUMPTION] until the format is confirmed), six numbered sections that mirror the rubric, figures and tables with captions, every statistic regenerated from code. Owner of assembly: G. Each section has one named author and one reviewer.

Section	Rubric points	Words	Must contain	Author / reviewer
Title, abstract, keywords	-	150	Question, data, method, key result with numbers, one limitation, research-prototype statement	A / G
1 Business understanding	4	400-500	Clinical context with citations; problem statement; RQ1-RQ5; success criteria S1-S9; scope and non-goals	A / G
2 Data understanding	4	500-700	Source and provenance (with unknowns marked), dimensions, dictionary table, types, units, key variables, initial limitations, lineage result	B / A
3 Preprocessing	5	700-900	Audit findings; treatment table (issue, diagnostic, decision, effect); views A-D with counts; leakage rules; integration/lineage; before/after table (Fig. 5)	C and D / B
4 EDA and statistics	6	900-1,200	Descriptives; Fig. 1-7 each interpreted; correlation analysis; hypothesis table H1-H5 for views A-D; assumptions; effect sizes with CIs; power; what the statistics do not show	E and F / D
5 Feature engineering	3	400-600	Feature spec table; encoding, scaling, transformation decisions with reasons; selection protocol and its honest outcome; final sets F0-F2	G / F
6 Conclusions and next steps	3	400-600	Findings per RQ; limitations; modelling plan (Ch. 14) without implementing; ethical note	A / G
References, appendix	(in 6)	-	Consistent citation style; reproducibility instructions (5 commands); link or checksum for the data	G / A

12.1 Writing rules

- **Claim, evidence, limitation.** Every paragraph in sections 3 to 6 states a claim, points to a figure or table number with the key number, and says what the evidence does not show.
- **Interpretation, not output.** The rubric says 'analytical discussion rather than raw outputs': never paste a statistic without saying what it means for a research question.
- **Numbers only from results.json.** Format: counts with thousands separators; proportions with one decimal as percentages; p-values in scientific notation when below 0.001; ORs with 95% CI.
- **Every figure** has a number, a caption that states the question and the takeaway, labelled axes with units, and is cited in the text before it appears.
- **Citations**: numbered or author-year, consistent; paraphrase sources, short quotes only; list the data source even if unknown ('provided by course; original provenance not documented').

Banned and approved phrasing

- Do not write: 'detect cancer', 'early detection', 'diagnose', 'accurate model', 'predicts cancer', 'proves', 'causes', 'clinically validated', 'the data is fake', 'AI doctor'.
- Write: 'recorded label', 'risk indication in a survey file', 'association in this file', 'consistent with', 'we found no evidence of', 'effect size', 'research prototype'.
- For the structure finding: 'statistically consistent with programmatic generation or resampling', never 'the data is synthetic' as a fact.

12.2 Draft abstract (update numbers from results.json before use)

Lung cancer is the leading cause of cancer death, and validated risk tools rely on detailed smoking history and age. We analyse a 22,457-row survey file of self-reported symptoms and habits with a recorded lung-cancer label, following the CRISP-DM process from business understanding to feature engineering. An audit found an ordered all-negative block of 1,455 rows containing only 103 distinct patterns, 2,573 exact duplicate rows, 133 label-conflicting patterns and an identifier that alone separates the labels (AUC 0.67). In the raw file all 14 binary features are associated with the label (Cramer's V up to 0.144; smoking odds ratio 1.49); after de-duplication and removal of the block, no association survives multiple-comparison correction (largest V 0.015; smoking odds ratio 1.00, 95% CI 0.92 to 1.08; smallest BH q 0.46). We specify leakage-safe preprocessing, three feature sets and a validation plan, and conclude that this file cannot support a predictive claim. This is a research prototype with no clinical use.

12.3 Table and figure inventory for the report

No.	Item	Source file
Tab. 1	Data dictionary (17 columns)	docs/data_dictionary.md
Tab. 2	Preprocessing treatments and effects	reports/tables/cleaning_log.csv
Tab. 3	Views A-D: n, positive rate	reports/tables/cleaning_log.csv
Tab. 4	Hypotheses H1-H5 across views	reports/tables/hypotheses.csv
Tab. 5	Feature specification and sets	docs/feature_spec.md
Fig. 1-7	Order artefact; prevalence by group; correlation; AGE; views; Cramer's V A vs D; smoking OR forest	reports/figures/*.png

13. Presentation blueprint (35 points)

20 minutes, seven speakers. The plan below totals **19:00** of speaking plus 0:30 for transitions, leaving 0:30 slack. Rubric weights: structure 6, methodological explanation 9, results and interpretation 9, Q and A 8, delivery and time 3. Each slide states **one message** and answers 'why did we choose this method?'. Planned work is labelled PLANNED on the slide.

#	Slide	Key message (say this first)	Visual	Speaker	Time	Cumulative
1	Title and one-line takeaway	Seven of us audited a lung-cancer survey file and found its signal is an artefact	Title, team, disclaimer	A	0:30	0:30
2	Why it matters	Real risk tools use age and smoking dose; survey flags are a weak substitute	USPSTF criteria; PLCOm2012 predictors	A	1:00	1:30
3	Problem, questions, success criteria	Five research questions; nine measurable criteria; no clinical claim	RQ table	A	1:00	2:30
4	The data	22,457 x 17, one file, no documentation; what each column is	Dictionary table, SHA-256	B	1:30	4:00
5	First quality picture and provenance	No nulls, but unknown source, label meaning and GENDER coding	Target bar (82% positive); lineage status	B	1:00	5:00
6	Audit I: the file is ordered	First 1,455 rows are all negative, 103 patterns, a different distribution	Fig. 1 and Fig. 2	C	1:30	6:30
7	Audit II: duplicates, conflicts, identifier	2,573 duplicates, 133 conflicting patterns, id alone gives AUC 0.67	Counts table	C	1:00	7:30
8	Cleaning policy and leakage rules	Four views A-D; minimal primary view; leakage prevented by construction	Fig. 5 and before/after table	D	2:00	9:30
9	EDA: structure of the cleaned data	Features look independent; AGE bell-shaped; positives and negatives alike	Fig. 4 and Fig. 3	E	2:00	11:30
10	Hypotheses and method	H1-H5 fixed in advance; effect sizes, CIs, BH correction, power	Hypothesis table	F	0:45	12:15
11	Results: the signal disappears	OR 1.49 raw vs 1.00 cleaned; V falls from 0.144 to 0.015; BH q at least 0.46	Fig. 6 and Fig. 7	F	2:15	14:30
12	Feature engineering	Domain-driven features; three sets; honest selection outcome	Feature spec table	G	2:30	17:00
13	Conclusions and limitations	RQs answered; file cannot support a predictive claim; specific limitations	RQ-answer table	A	1:00	18:00
14	Next steps: modelling plan	Leakage-safe validation, metrics, better data; model only as planned work	Plan timeline	A	1:00	19:00

13.1 Slide-by-slide notes

Slide	Delivery notes
2	Open with the USPSTF rule (50-80 years, 20 pack-years, quit within 15 years) and the PLCOm2012 predictor list. Land the point: our file has a yes/no smoking flag, no dose, no quit time. Therefore the honest question is about information content, not early detection.
3	Read the five RQs aloud in one sentence each. Show S1 to S9 as a table; point at S3 (reproducible) and S6 (leakage).
4	Show the dictionary as one compact table. Say explicitly: GENDER is a code, 1/2 flags recoded to 0/1. Mention the SHA-256 and that raw data is read-only.
5	Say what we do not know: source, licence, label meaning. This earns credibility and prevents the instructor from finding it first.
6	Walk Fig. 1 left to right: 'positive rate is zero for the first 1,455 rows, then jumps'. Then Fig. 2: 'later negatives and later positives are the same; this block is different'. One sentence: it behaves like another source.
7	Three numbers: 2,573, 133, 0.67. Explain pseudo-replication and why the identifier is a leak in one sentence each.
8	Show the four-view bar chart and the before/after table. State the rule: deduplicate first, shuffle, fit transforms on training only, no identifier. Say why B is primary.
9	Show the correlation heatmap with its clipped colour scale: 'the clipped scale shows nothing, which is the finding: max |r| = 0.022'. AGE bell-shaped, skew -0.01.
10	Present H1 as the primary pre-registered hypothesis and why smoking: domain prior. State alpha, the tests, the BH rule, that effect sizes and CIs come with every p.
11	Fig. 7 first (OR 1.49 to 1.00). Then Fig. 6 (V falls). Then the interpretation: not 'smoking is harmless' but 'this file cannot show it'. State the power sentence: effects above about 1.3 points would have been detected.
12	Table of features with the justification column; verify() passes. Honest sentence: selection by evidence is inconclusive because nothing is informative; domain priors define F2.
13	One row per RQ with the answer. Limitations: provenance, label meaning, no dose, single file.
14	Planned (labelled): grouped stratified nested CV, PR-AUC and calibration, subgroup checks, negative-control expectation (chance-level AUC), request better data. End on the research-prototype statement.

13.2 Slide design rules

- One message per slide, headline is the message. At least 20 pt body text; figures fill at least 60% of the slide area; no paragraph longer than two lines.
- Palette consistent with the handbook figures; every slide has the footer 'Research prototype - not a diagnostic tool' and a slide number.
- Every number on a slide appears in results.json. Cite sources on the slide (author, year) and give the full list on the last slide.
- **Backup slides** (hidden, after slide 14): view sensitivity table; family-test table; power calculation; PCA scree; leakage taxonomy mapping; lineage check result; modelling plan detail; glossary.

13.3 Rehearsal protocol

Run	When	Rules	Pass criterion
1	Mon 26 evening	Full run, stopwatch, no stopping, someone notes unclear sentences	Under 21:00; list of fixes
2	Tue 27 midday	Same, after fixes; handovers rehearsed ('Over to B, who will show...')	Under 19:30
Q and A 1	Tue 27 morning	Questions from the bank assigned to the partner of the author; 90 seconds per answer	Each member at least 80% of answers acceptable
Q and A 2	Tue 27 afternoon	Surprise questions from A and from a volunteer outside the team	Weak topics identified and taught back

Delivery. Speak to the audience, not the slide; look at the instructor; describe each figure's axes before its message; use 'we' for team work and 'I' for what you personally implemented; if you do not know, say what you would check and how.

13.4 Q and A bank: 35 questions with model answers

Answer in 60 to 90 seconds: direct answer, one number or reason, one limitation. Numbers are VERIFIED unless stated. The 'Who' column says who answers first in practice (partners rotate).

#	Who	Question	Model answer
1	A	Why not just train a model and report accuracy?	The rubric excludes training, and more importantly accuracy is uninformative: always answering 'positive' scores 82% (87% after cleaning). The raw file's apparent signal is an artefact of duplicates and an ordered block, so any score would measure leakage. We specify a leakage-safe evaluation instead (Ch. 14).
2	A	What is your clinical claim?	None. It is a research prototype. Real screening eligibility uses age, pack-years and quit time (USPSTF 2021); our file has none of the last two.
3	A	Why an integrity-first approach?	Evidence led us there: the audit showed the raw associations vanish after treatment. This is the CRISP-DM feedback loop: phase 3 changed phase 1.
4	A	What would change your conclusion?	Provenance showing the label is clinical and the file genuine; a match of our rows to the original survey that justifies the leading block; a correlation structure inside the later rows. Any of these would change interpretation, not our counts.
5	B	Where does the data come from?	Unknown to us: the ZIP's README has only a title. The schema matches a public 309-row Kaggle survey collected from an online prediction system (the page itself says 284 instances). Our file adds an id and 22,000 more rows. The lineage check is task B-4 and its status is on the slide.
6	B	What does GENDER = 1 mean?	We do not know; the source survey used M/F. We report it as GENDER_CODE and make no statement about sex.
7	B	Why are flags coded 1 and 2?	Inherited from the source survey (YES = 2, NO = 1). We recode to 0/1 so that means are shares of YES; the raw file is untouched.
8	B	Are there missing values?	No nulls and no sentinel codes: symptom columns hold only 1 and 2, AGE is 21 to 87, ids are 1 to 22,457 unique. If there were, we would diagnose the mechanism (MCAR, MAR, MNAR) and fit any imputer on training data only.
9	C	Duplicates could be different real people. Why remove them?	With 14 binary answers and age, chance collisions are expected about 600 times in rows from 1,455 on, but we observe 1,352, and the leading block collapses from 1,455 rows to 103 patterns. We cannot tell which copies are real, so we keep one copy (view B) and show that views C and D give the same conclusion.
10	C	Why keep label-conflicting rows in the primary view?	Minimal intervention: removing them is a judgement about label noise. View C removes them (19,618 rows) and the odds ratio is 1.04 (0.96 to 1.14), so the conclusion is unchanged.
11	C	How do you know the first 1,455 rows are a different source?	All negative, 103 distinct patterns repeated up to 86 times, smokers 29% vs 56% in later negatives, Cramer's V 0.18 to 0.54 on every feature, ages down to 21 while the rest starts at 30. We say 'behaves like a separate source', not that it is one.
12	D	Why not remove AGE outliers?	IQR fences 40.5 to 84.5 flag 221 rows, but 21 to 87 are plausible ages and there are no impossible values. Removing them would be arbitrary; view D already drops the young block as a sensitivity check.
13	D	How did you avoid leakage in preprocessing?	Rules L1 to L6: deduplicate and group before splitting, shuffle with seed, fit learned transforms on training only, exclude the identifier, nest selection in CV later, log all counts. Tests assert the identifier is absent and the scaler is fitted on its input only.
14	D	Why not impute? What if values were missing?	Nothing is missing. In general: diagnose the mechanism, add missing-indicators if informative, impute with a model fitted inside training folds, and run a sensitivity analysis with complete cases.
15	D	Is patient_id really a leak?	Yes: because rows are ordered with an all-negative block first, id alone separates classes with AUC 0.67. In a real deployment the id would carry nothing; that is exactly why it is illegitimate.
16	E	Your heatmap is blank. Why show it?	The blank is the evidence: max |r| is 0.022 and 4.4% of 91 pairwise tests reject at 5% (the chance rate). Real survey data usually show correlated answers, so this informs provenance and tells us PCA and redundancy filtering have nothing to find.
17	E	Why is AGE bell-shaped, is that a problem?	Skewness -0.006 and excess kurtosis -0.055: almost perfectly normal, mean 62.2, sd 8.2. Real clinical ages are rarely this symmetric. It is a clue about generation, not a problem for analysis; no transformation needed.
18	E	What do positives and negatives look like?	After row 1,455 they are practically identical on every feature (e.g. smokers 56.0% vs 56.2%). The leading-block negatives differ strongly from both.
19	F	What does p = 4.5e-30 mean in the raw file?	If smoking were unrelated to the label, a gap like 84.8% vs 78.9% would be extremely surprising. But the gap arises from the leading block; the effect size is small (V 0.076) and disappears after treatment (OR 1.00). The p-value does not measure importance.

#	Who	Question	Model answer
20	F	Why Cramer's V and odds ratios?	With n of about 20,000 almost any difference is significant, so effect sizes and intervals carry the information. V is a 0 to 1 effect size for chi-square; the OR with CI is interpretable for a 2x2 table.
21	F	Why a chi-square test for H1 and not logistic regression?	Two binary variables: a 2x2 chi-square is the direct test, the OR comes from the table, and a permutation test gives a distribution-free check (p = 0.51). A joint likelihood-ratio test of all features is the multivariable complement (p = 0.31 in view D).
22	F	Why Benjamini-Hochberg?	14 simultaneous tests give about a 51% chance of at least one false positive at 5%. BH controls the false discovery rate and is less conservative than Bonferroni. Smallest raw p 0.033 becomes q = 0.46.
23	F	What assumptions does the chi-square test make?	Independent observations (hence deduplication) and expected counts at least 5, trivially met here. Independence is the fragile one: copies would violate it.
24	F	Why Mann-Whitney for AGE?	It compares ranks and does not assume normality; AGE in the bulk is nearly normal so a Welch t-test would agree, but the leading block makes the raw distribution non-normal. In view D p = 0.051 and CLES = 0.512.
25	F	What does CLES = 0.512 mean?	The probability that a random positive is older than a random negative: 51.2% versus 50% by chance. A negligible difference of about 0.3 years.
26	F	What is the power of your test?	In view D the minimum detectable risk difference for smoking is about 1.3 percentage points at 80% power. A true effect larger than that would very likely be detected.
27	F	Does a null result mean smoking is not a risk factor?	No. It means this file does not contain that relationship, which itself questions the file's construct validity, because real populations show a strong smoking effect (USPSTF).
28	G	Why create symptom counts?	Domain grouping: four respiratory symptoms (0 to 4) reduces sparse flags to a burden score. We tested association: rho = -0.002 in view D, so it adds no information here; it is retained as a candidate for modelling comparisons.
29	G	Why z-score AGE?	Scale-sensitive models (regularised regression, SVM, neural nets) need comparable scales. The mean and sd are learned parameters, so they must come from training folds only.
30	G	Why no log transformation?	Transformations fix skew; AGE skewness is -0.006. Applying one would be unjustified.
31	G	Why did you not drop features?	No evidence-based reason: VIF is 1.00 to 1.03, no near-constant flags, and no feature is informative to rank. Dropping by p-value on all data would also leak. We use a domain-driven set F2 and keep F0 and F1 for comparison.
32	G	Why not PCA?	The PCA spectrum is flat: the first component explains 7.0% against 6.7% for no structure. Extraction cannot compress what is not correlated.
33	A	What models will you build and how will you evaluate them?	PLANNED: majority and prevalence baselines, regularised logistic regression, a gradient-boosted tree model; stratified, grouped (by feature pattern), nested CV; PR-AUC, ROC-AUC with bootstrap CIs, calibration, subgroup results; on this file we expect chance-level performance and treat it as a negative control for our pipeline.
34	A	What if a model reaches 95% accuracy on the raw file?	We would assume leakage first: duplicates, the identifier, the block, or an illegitimate feature. We would repeat with grouped folds on view B or D and expect the score to collapse.
35	A	How would you get better data and how generalisable are your results?	Our findings concern this file only. Better options: the original survey with documented provenance, or controlled-access trial data such as NLST or PLCO via NCI CDAS, or public CT data; each needs approval or heavier compute. We would choose by a decision matrix (Ch. 14).

14. After the midterm: responsible modelling plan

[PLANNED]

None of this is implemented for the midterm (D-05). It is the plan we will show on slide 14 and refine in the first week after the midterm. Semester dates beyond October 28 are unknown **[OPEN]**, so the schedule is relative.

14.1 Evaluation protocol (specified before any model is trained)

Element	Specification	Reason
Data view	Primary B; sensitivity C and D; A only as a leakage demonstration	Matches D-03
Split	Shuffle (seed 42), then repeated stratified **grouped** k-fold (groups = identical feature pattern), 5 x 5 repeats; hold out a final 20% test set touched once	Prevents duplicate leakage; gives variance estimates
Preprocessing	scikit-learn Pipeline: recode, scaler, optional selector, model, all fitted inside folds	L3 and L5
Baselines	Majority class; prevalence-matched random; smoking-only rule; age 50-80 and smoker rule (USPSTF proxy); patient_id-only model as a leakage detector	Anything we build must beat trivial and domain baselines
Models	L2-regularised logistic regression; L1/elastic-net logistic regression; gradient-boosted trees; (optional) small MLP. Tune inside nested CV with a fixed search budget	Interpretable first, flexible second; avoid tuning on test data
Metrics	ROC-AUC and PR-AUC with bootstrap 95% CIs; balanced accuracy; MCC; sensitivity and specificity at a pre-stated threshold; Brier score; calibration slope and intercept	Imbalance and risk estimation need more than accuracy
Calibration and utility	Reliability curve; isotonic or Platt calibration inside folds; decision-curve analysis	Risk indications must be calibrated to be usable
Interpretability	Coefficients with CIs; permutation importance on held-out folds; partial dependence; SHAP only as supplementary and with a stability check	Explain without over-interpreting
Uncertainty and robustness	Bootstrap CIs; label-noise injection (flip 5%, 10%); leave-one-feature-out; train on B and test on D and C; conformal prediction as an extension	Does the conclusion survive perturbation?
Subgroups	GENDER_CODE and age bands (with n and CIs); report that no ethnicity or socioeconomic variable exists	Fairness within data limits
Reporting	TRIPOD+AI checklist items mapped to report sections; leakage info sheet	Transparent reporting standard

14.2 Negative controls: our pipeline's truth serum

Because the cleaned data appear to contain no signal, a correct pipeline must return **chance-level performance** on it. Each control below has an expected result; a deviation means a bug or a leak, which is itself a finding.

Control	Expected result	If violated
Permute the labels (view B) and run the full pipeline	ROC-AUC about 0.50, PR-AUC about prevalence (0.87)	Leak inside the pipeline
Model on patient_id alone, raw file, shuffled CV	AUC about 0.67	Confirms the identifier shortcut; on B and D it should fall to about 0.50
Raw file, shuffled CV, duplicates kept	AUC about 0.67 to 0.70 (inflated)	Documents the size of duplicate and block inflation
View B, shuffled stratified CV	AUC about 0.50	Signal exists that our tests missed: investigate
Train on A, test on D	AUC at or below chance	Shows the A-trained model learned the block

14.3 Choosing the data track for the rest of the semester [JUDGMENT]

Option	Scientific validity for a lung-cancer claim (w 0.30)	Feasibility in the time left (w 0.25)	Access lead time (w 0.20)	Compute needs (w 0.10)	Novelty and impact (w 0.15)	Weighted score
1 Current file as methodological negative control	2	5	5	5	3	3.80
2 Original public survey (309 rows) for lineage and comparison	2	5	4	5	2	3.45
3 NCI CDAS controlled-access trial data (PLCO or NLST) with a PLCOm2012-style baseline	5	3	1	5	4	3.55
4 Public CT imaging (for example LIDC-IDRI) with a CNN	4	2	4	2	5	3.45
5 Single-hospital cohort (P4-LUCAT)	3	2	1	5	3	2.55

Scores 1 (poor) to 5 (excellent) are the handbook author's judgment, not measurements; change weights to reflect your priorities. P4-LUCAT covers diagnosis and treatment of known patients, which does not match a risk-indication question, hence its low validity.

> **Recommendation (JUDGMENT)**
>
> Keep option 1 as the methodological backbone: the audit and negative-control pipeline is already the project's distinctive contribution. In parallel, ask the instructor about access to better data and, if permitted, start the NCI CDAS application now because it has the longest lead time (verify the current procedure on the CDAS site). Treat imaging as a stretch goal only if GPU access is confirmed. Decide at the Week 1 checkpoint.

14.4 Relative schedule after the midterm [ASSUMPTION]

Week	Goal	Deliverable
1	Debrief; decide data track; lineage result; finalise evaluation protocol and baselines	Protocol document (frozen); decision log update
2	Implement pipeline, baselines and negative controls	Controls reproduce expected values
3	Candidate models with nested CV; calibration; bootstrap CIs	Results tables with CIs
4	Robustness, subgroup analysis, error analysis, interpretability	Robustness and fairness section
5	Second data track work (if approved) or extended negative-control study; TRIPOD+AI mapping	Draft final report
6	Final report, presentation, reproducibility package, release tag	Submission

15. Landscape of existing projects: inspire, evaluate, consider

Research performed on 24 October 2026 with web search. Entries reflect what the public pages state; we could not open every full paper, so evaluations concern what is documented. **[SOURCED]** Our judgments are marked.

Project / source	Data and task	Reported approach and claims	Strengths	Weaknesses and risks [JUDGMENT]	Take-away for us
Kaggle 'Lung Cancer' survey (user mysarah madbhat)	309-row table (16 columns); description says 284 instances; collected from an online lung-cancer prediction system; YES = 2, NO = 1	Public dataset for classification	Small, simple, widely reused	Label origin unclear; duplicate rows; tiny n; documentation inconsistent	Likely ancestor of our schema; verify lineage (B-4)
Kaggle 'Lung Cancer Dataset' (user akashnath29)	3,000 rows, same symptom columns	Prose mentions age at diagnosis, smoking categories, vital signs	Larger	Description does not match columns	Do not trust documentation without checking columns
RespiraAI (Hugging Face Space)	309-row survey; predict cancer; maximise recall	Drops duplicates, maps YES/NO to 1/0, removes AGE for 'multicollinearity'; published as a Hugging Face Space	Deployed demo; recall-oriented motivation; deduplicates	Says negatives outnumber positives, contradicting other write-ups; AGE removal not supported by our VIF of 1.00; no external validation	Good packaging; emulate dedup and a demo later; verify every claim
IJERT 'Predicting Lung Cancer Among Smokers'	Same 309-row survey; label read as 'suspected of having lung cancer'	Label encoding, ML classifiers; says data 'balanced' and without duplicates	Candid about label meaning	Balanced/duplicate claims conflict with other sources; small n	Label semantics matter; always compute class balance yourself
'Lung cancer detection and classification using ML algorithms' (authors not verified)	Survey data; SVM, Naive Bayes, KNN, decision tree, random forest	Reports decision tree with 100% accuracy	Compares several models	100% accuracy on tiny survey data is a leakage or triviality red flag; 'detection' framing	Exactly the pattern our leakage controls guard against
PLCOm2012 (Tammemagi et al., NEJM 2013)	80,375 ever-smokers from the PLCO trial; 6-year lung-cancer risk	Logistic model using, among others, age, education, BMI, family history, COPD, smoking status, intensity, duration and quit time	Large cohort, validated, beat NLST criteria (41.3% fewer cancers missed)	Needs detailed variables; US trial population	Gold-standard tabular baseline; shows what variables matter
Sybil (Mikhael et al., JCO 2023)	Single low-dose CT; NLST for training; validated on MGH and CGMH	1-year ROC-AUC 0.92, 0.86, 0.94; 6-year C-index 0.75, 0.81, 0.80; public release reported by the authors	External validation in two countries; authors report a public release	Trained on smokers only; needs prospective study	Benchmark for rigour: external validation, CIs, openness
Early detection in the NLST dataset (preprint, 2023)	NLST CT; two-stage model on 620 cancer patients plus controls	Predict future cancer from lung CT	Uses the major screening trial; discusses false positives (over 25% positive screens, under 10% cancer)	Preprint; clinical data limited	Context for false-positive cost

Project / source	Data and task	Reported approach and claims	Strengths	Weaknesses and risks [JUDGMENT]	Take-away for us
Multimodal LDCT and plasma proteomics dataset (medRxiv preprint, 2025)	67 cancers, 68 benign-nodule and 71 no-nodule controls, 5 false positives; matched by sex, age, timing	Open-access research resource	Matched design; multimodal	Small; preprint	Example of a careful design (matching) we could cite as future data
P4-LUCAT (DOAJ-indexed data descriptor)	1,242 lung-cancer patients, Madrid hospital: diagnosis, treatment, adverse effects	Open dataset description	Real clinical records	Treatment-focused; not a risk-indication task	Shows what real clinical data look like (correlated, messy)
Medical AI for early lung-cancer detection: a survey (arXiv 2410.14769)	Review of CT-based nodule detection, segmentation, classification	Taxonomy of methods; companion GitHub list	Broad map of the field	Review, not a method	Cite for context; shows imaging dominates serious work

15.1 What the best work has in common, and how we position ourselves

Practice of strong work (PLCOm2012, Sybil)	Common in weak write-ups	What we do
Clinically meaningful variables (dose, time since quitting) or imaging	A handful of yes/no flags	State the gap explicitly; do not claim detection
External validation in independent cohorts	Single random split, often with duplicates	Plan grouped nested CV; document that no external data exist
Confidence intervals and calibration	Single accuracy figure	Effect sizes with CIs now; calibration later
Transparent reporting and open artefacts	No data card, no seeds	Data card, checksums, tests, TRIPOD+AI mapping
Comparison with existing criteria (NLST, USPSTF)	No baselines	Domain baselines (age and smoking rule) in the plan
Honest limitations	Generic disclaimers	Evidence-based limitations in Ch. 6, phase 6

Inspiration to carry into the deck

Sybil is remembered for external validation and openness, PLCOm2012 for beating the guideline with a few well-chosen variables. Our distinctive claim is different but equally rigorous: **we measured when a dataset cannot support a model, and we show how.** Put one slide-14 line on each: 'baseline against USPSTF-style rule', 'external validation required', 'TRIPOD+AI reporting'.

16. References

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Appendix A. Integrity audit script (the exact script that produced Chapter 4)

Save as `audit_all.py` and run `python audit_all.py data/raw/lung_cancer_combined_22457.csv`. Requires pandas, numpy, scipy, statsmodels. Compare your output with the listing that follows the code.

```
"""audit.py - Reproducible integrity audit of lung_cancer_combined_22457.csv (read-only).
Usage: python audit.py path/to/lung_cancer_combined_22457.csv
Requires: pandas numpy scipy statsmodels"""
import sys, numpy as np, pandas as pd
from scipy import stats
from statsmodels.stats.multitest import multipletests

PATH = sys.argv[1]
T, ID = "LUNG_CANCER", "patient_id"
df = pd.read_csv(PATH)
X = [c for c in df.columns if c not in (ID, T)]
BIN = [c for c in X if c != "AGE"]
rng = np.random.default_rng(42)
print(f"[1] shape={df.shape} nulls={int(df.isna().sum().sum())} id_unique={df[ID].is_unique}")
print(f"[2] target counts={df[T].value_counts().to_dict()} pos_rate={df[T].mean():.4f}")

# [3] row-order artefact
fp = int(np.argmax(df[T].values == 1))
print(f"[3] first positive at row index {fp}; AUC(patient_id->target)="
      f"{stats.mannwhitneyu(df.loc[df[T]==1,ID], df.loc[df[T]==0,ID]).statistic/((df[T]==1).sum()*(df[T]==0).sum()):.3f}")

# [4] duplicates and label conflicts
dup = df.duplicated(subset=X + [T])
g = df.groupby(X)[T].agg(["nunique", "size"])
print(f"[4] exact dup rows (features+label)={int(dup.sum())}; dup on features only={int(df.duplicated(subset=X).sum())}; "
      f"distinct patterns={len(g)}; conflicting patterns={int((g['nunique']>1).sum())} "
      f"({int(g.loc[g['nunique']>1,'size'].sum())} rows); max multiplicity={int(g['size'].max())}")

# [5] leading block vs later negatives (distribution shift = source effect)
lead, rest = df.iloc[:fp], df.iloc[fp:]
neg_rest = rest[rest[T] == 0]
print(f"[5] lead block n={len(lead)}, distinct patterns={len(set(map(tuple, lead[X].values)))}, all negative={lead[T].eq(0).all()}")
vs = []
for c in BIN:
    tab = pd.crosstab(np.r_[np.zeros(len(lead)), np.ones(len(neg_rest))], np.r_[lead[c], neg_rest[c]])
    chi2 = stats.chi2_contingency(tab, correction=False)[0]
    vs.append(np.sqrt(chi2 / tab.values.sum()))
print(f"    Cramer's V lead-negatives vs later-negatives: min={min(vs):.2f} max={max(vs):.2f} (all p<1e-25)")

# [6] bulk behaves like independent columns?
bulk = rest.drop_duplicates(subset=X + [T])
cm = np.abs(bulk[X].corr().values[~np.eye(len(X), dtype=bool)])
sim_d = np.mean([pd.DataFrame({c: rng.choice(rest[c].values, len(rest)) for c in X}).duplicated().sum() for _ in range(10)])
print(f"[6] bulk (rows>={fp}) n={len(rest)}; max|pairwise corr| after dedup={cm.max():.3f}; "
      f"dup rows observed={int(rest.duplicated(subset=X).sum())} vs expected under independence ~{sim_d:.0f}")

# [7] cleaning policies
key = df[X].astype(str).agg("|".join, axis=1)
conf_keys = set(key[df.set_index(X).index.isin(g[g["nunique"] > 1].index)])
A = df
B = df[~dup]
C = B[~key[B.index].isin(conf_keys)]
D = B[B.index >= fp]
views = {"A raw": A, "B dedup": B, "C dedup-no-conflict": C, "D dedup-no-lead-block": D}
print("[7] policy table:")
for k, d in views.items():
    print(f"    {k:22s} n={len(d):6d} pos_rate={d[T].mean():.4f}")
```

```
# [8] primary hypothesis H1: SMOKING (2=yes) vs target, per view
def h1(d, col="SMOKING"):
    a, b = d[d[col] == 2], d[d[col] == 1]
    p1, p0 = a[T].mean(), b[T].mean()
    tab = np.array([[a[T].sum(), len(a) - a[T].sum()], [b[T].sum(), len(b) - b[T].sum()]])
    chi2, p, _, _ = stats.chi2_contingency(tab, correction=False)
    or_ = (tab[0, 0] * tab[1, 1]) / (tab[0, 1] * tab[1, 0])
    se = np.sqrt((1 / tab).sum()); lo, hi = np.exp(np.log(or_) + np.array([-1.96, 1.96]) * se)
    return p1, p0, p1 - p0, or_, lo, hi, p, np.sqrt(chi2 / len(d))
print("[8] H1 smoking->cancer: view | P(pos|smoker) P(pos|non) diff OR[95%CI] chi2-p V")
for k, d in views.items():
    p1, p0, df_, o, lo, hi, p, v = h1(d)
    print(f"    {k:22s} {p1:.3f} {p0:.3f} {df_:+.3f} OR={o:.2f}[{lo:.2f},{hi:.2f}] p={p:.2g} V={v:.3f}")

# [9] all 14 binary features on view D, BH-FDR; plus AGE
rows = []
for c in BIN:
    tab = pd.crosstab(D[c], D[T]); chi2, p, _, _ = stats.chi2_contingency(tab, correction=False)
    rows.append((c, np.sqrt(chi2 / len(D)), p))
r = pd.DataFrame(rows, columns=["feature", "V", "p"]); r["q_BH"] = multipletests(r.p, method="fdr_bh")[1]
print("[9] view D, chi-square per feature (sorted by p):")
print(r.sort_values("p").round(4).to_string(index=False))
mw = stats.mannwhitneyu(D[D[T] == 1].AGE, D[D[T] == 0].AGE)
print(f"    AGE Mann-Whitney p={mw.pvalue:.3f}, CLES={mw.statistic/((D[T]==1).sum()*(D[T]==0).sum()):.3f}")

# [10] power: minimum detectable difference (80% power, alpha .05) on view D for a balanced-ish binary feature
n1 = (D["SMOKING"] == 2).sum(); n0 = (D["SMOKING"] == 1).sum(); pb = D[T].mean()
mde = (1.96 + 0.8416) * np.sqrt(pb * (1 - pb) * (1 / n1 + 1 / n0))
print(f"[10] view D smoker n={n1}, non-smoker n={n0}: minimum detectable risk difference ~ {mde:.3f}")
```

Expected output (produced 24 Oct 2026)

```
[1] shape=(22457, 17) nulls=0 id_unique=True
[2] target counts={1: 18434, 0: 4023} pos_rate=0.8209
[3] first positive at row index 1455; AUC(patient_id->target)=0.670
[4] exact dup rows (features+label)=2573; dup on features only=2706; distinct patterns=19751; conflicting patterns=133 (315 rows); max multiplicity=86
[5] lead block n=1455, distinct patterns=103, all negative=True
    Cramer's V lead-negatives vs later-negatives: min=0.18 max=0.54 (all p<1e-25)
[6] bulk (rows>=1455) n=21002; max|pairwise corr| after dedup=0.022; dup rows observed=1352 vs expected under independence ~594
[7] policy table:
    A raw                  n= 22457 pos_rate=0.8209
    B dedup                n= 19884 pos_rate=0.8661
    C dedup-no-conflict    n= 19618 pos_rate=0.8711
    D dedup-no-lead-block  n= 19781 pos_rate=0.8706
[8] H1 smoking->cancer: view | P(pos|smoker) P(pos|non) diff OR[95%CI] chi2-p V
    A raw                  0.848 0.789 +0.059 OR=1.49[1.39,1.59] p=4.5e-30 V=0.076
    B dedup                0.868 0.864 +0.003 OR=1.03[0.95,1.12] p=0.52 V=0.005
    C dedup-no-conflict    0.873 0.868 +0.005 OR=1.04[0.96,1.14] p=0.3 V=0.007
    D dedup-no-lead-block  0.870 0.871 -0.001 OR=1.00[0.92,1.08] p=0.91 V=0.001
[9] view D, chi-square per feature (sorted by p):
             feature      V      p   q_BH
              GENDER 0.0152 0.0327 0.4583
          CHEST_PAIN 0.0106 0.1349 0.7781
  SHORTNESS_OF_BREATH 0.0096 0.1748 0.7781
    ALCOHOL_CONSUMING 0.0086 0.2261 0.7781
SWALLOWING_DIFFICULTY 0.0068 0.3383 0.7781
             FATIGUE 0.0065 0.3633 0.7781
             ANXIETY 0.0051 0.4758 0.7781
             ALLERGY 0.0047 0.5093 0.7781
       YELLOW_FINGERS 0.0043 0.5490 0.7781
            WHEEZING 0.0042 0.5558 0.7781
     CHRONIC_DISEASE 0.0028 0.6952 0.8815
            COUGHING 0.0017 0.8058 0.8815
       PEER_PRESSURE 0.0016 0.8185 0.8815
             SMOKING 0.0008 0.9110 0.9110
    AGE Mann-Whitney p=0.051, CLES=0.512
[10] view D smoker n=11080, non-smoker n=8701: minimum detectable risk difference ~ 0.013
```

Appendix B. Reference implementation (tested: 6 pytest tests pass)

Use this **after** attempting each task yourself: the goal is to master the concepts by implementing them. Compare outputs, not code style. Place files as shown in Ch. 8.

src/config.py

```
"""Single source of truth for paths, constants and the expected raw-file fingerprint."""
from pathlib import Path

ROOT = Path(__file__).resolve().parents[1]          # repo root; no hard-coded absolute paths
RAW = ROOT / "data" / "raw" / "lung_cancer_combined_22457.csv"
INTERIM, PROCESSED = ROOT / "data" / "interim", ROOT / "data" / "processed"
FIG, TAB = ROOT / "reports" / "figures", ROOT / "reports" / "tables"

TARGET, ID, SEED = "LUNG_CANCER", "patient_id", 42
SYMPTOMS_RESP = ["COUGHING", "WHEEZING", "SHORTNESS_OF_BREATH", "CHEST_PAIN"]
SYMPTOMS_OTHER = ["FATIGUE", "SWALLOWING_DIFFICULTY"]
EXPECTED = {"rows": 22457, "cols": 17, "sha256_prefix": "4ad2333067fcb0a9"}
```

src/data_io.py

```
"""Load the raw file read-only, verify its fingerprint and validate the schema."""
import hashlib
import pandas as pd
from config import RAW, EXPECTED, TARGET, ID

BINARY_12 = ["SMOKING", "YELLOW_FINGERS", "ANXIETY", "PEER_PRESSURE", "CHRONIC_DISEASE", "FATIGUE", "ALLERGY",
             "WHEEZING", "ALCOHOL_CONSUMING", "COUGHING", "SHORTNESS_OF_BREATH", "SWALLOWING_DIFFICULTY", "CHEST_PAIN"]
FEATURES = ["GENDER", "AGE"] + BINARY_12


def sha256(path) -> str:
    h = hashlib.sha256()
    with open(path, "rb") as f:
        for chunk in iter(lambda: f.read(1 << 20), b""):
            h.update(chunk)
    return h.hexdigest()


def validate_schema(df: pd.DataFrame) -> None:
    assert list(df.columns) == [ID] + FEATURES + [TARGET], "unexpected columns/order"
    assert df.shape == (EXPECTED["rows"], EXPECTED["cols"]), f"unexpected shape {df.shape}"
    assert df.isna().sum().sum() == 0, "missing values found"
    assert (df.dtypes == "int64").all(), "non-integer dtype found"
    assert df[ID].is_unique, "patient_id not unique"
    assert set(df["GENDER"]) <= {0, 1} and set(df[TARGET]) <= {0, 1}
    assert all(set(df[c]) <= {1, 2} for c in BINARY_12), "binary flags must be coded 1/2"
    assert df["AGE"].between(18, 110).all(), "implausible AGE"


def load_raw(verify: bool = True) -> pd.DataFrame:
    if verify:
        assert sha256(RAW).startswith(EXPECTED["sha256_prefix"]), "RAW FILE CHANGED - restore from the original ZIP"
    df = pd.read_csv(RAW)
    validate_schema(df)
    return df
```

src/audit.py

```
"""Integrity audit: order artefact, duplicates, label conflicts, source shift. Read-only."""
import json
import numpy as np
import pandas as pd
from scipy import stats
from config import TARGET, ID, TAB
from data_io import load_raw, FEATURES

BIN = [c for c in FEATURES if c != "AGE"]


def leading_block_end(df: pd.DataFrame) -> int:
    """Index of the first positive label = length of the all-negative leading block."""
    return int(np.argmax(df[TARGET].values == 1))


def duplicate_report(df: pd.DataFrame) -> dict:
    dup_exact = df.duplicated(subset=FEATURES + [TARGET])
    g = df.groupby(FEATURES)[TARGET].agg(["nunique", "size"])
    return {"dup_rows_features_and_label": int(dup_exact.sum()),
            "dup_rows_features_only": int(df.duplicated(subset=FEATURES).sum()),
            "distinct_patterns": int(len(g)),
            "patterns_seen_more_than_once": int((g["size"] > 1).sum()),
            "max_multiplicity": int(g["size"].max()),
            "conflicting_patterns": int((g["nunique"] > 1).sum()),
            "rows_in_conflicting_patterns": int(g.loc[g["nunique"] > 1, "size"].sum())}


def source_shift(df: pd.DataFrame, fp: int) -> pd.DataFrame:
    """Cramer's V between leading-block negatives and later negatives, per binary feature."""
    lead, later = df.iloc[:fp], df.iloc[fp:]
    later = later[later[TARGET] == 0]
    rows = []
    for c in BIN:
        tab = pd.crosstab(np.r_[np.zeros(len(lead)), np.ones(len(later))], np.r_[lead[c], later[c]])
        chi2, p, _, _ = stats.chi2_contingency(tab, correction=False)
        rows.append((c, float(np.sqrt(chi2 / tab.values.sum())), float(p)))
    return pd.DataFrame(rows, columns=["feature", "cramers_v", "p"]).sort_values("cramers_v", ascending=False)


def auc_of_id(df: pd.DataFrame) -> float:
    """How well does the row identifier alone separate the classes? (should be ~0.5 if no leakage)"""
    pos, neg = df.loc[df[TARGET] == 1, ID], df.loc[df[TARGET] == 0, ID]
    return float(stats.mannwhitneyu(pos, neg).statistic / (len(pos) * len(neg)))


if __name__ == "__main__":
    df = load_raw()
    fp = leading_block_end(df)
    out = {"first_positive_index": fp,
           "leading_block_distinct_patterns": int(len(df.iloc[:fp][FEATURES].drop_duplicates())),
           "auc_patient_id_alone": round(auc_of_id(df), 3), **duplicate_report(df)}
    TAB.mkdir(parents=True, exist_ok=True)
    (TAB / "audit_summary.json").write_text(json.dumps(out, indent=2))
    source_shift(df, fp).round(4).to_csv(TAB / "source_shift.csv", index=False)
    print(json.dumps(out, indent=2))
```

src/clean.py

```
"""Build the four analysis views A-D with an auditable before/after log. Raw data is never modified."""
import pandas as pd
from config import TARGET, INTERIM, TAB
from data_io import load_raw, FEATURES
from audit import leading_block_end


def make_views(df: pd.DataFrame) -> dict:
    fp = leading_block_end(df)
    is_dup = df.duplicated(subset=FEATURES + [TARGET], keep="first")
    B = df[~is_dup]                                                  # exact duplicates removed
    n_labels = df.groupby(FEATURES)[TARGET].transform("nunique")     # >1 means conflicting labels
    C = B[n_labels[B.index] == 1]                                    # B minus label-conflict rows
    D = B[B.index >= fp]                                             # B minus the leading block
    return {"A_raw": df, "B_dedup": B, "C_dedup_noconflict": C, "D_dedup_nolead": D}


def summarize(views: dict) -> pd.DataFrame:
    rows = [{"view": k, "n": len(v), "positives": int(v[TARGET].sum()), "pos_rate": round(float(v[TARGET].mean()), 4),
             "min_age": int(v.AGE.min()), "mean_age": round(float(v.AGE.mean()), 2)} for k, v in views.items()]
    return pd.DataFrame(rows)


if __name__ == "__main__":
    df = load_raw()
    views = make_views(df)
    INTERIM.mkdir(parents=True, exist_ok=True)
    TAB.mkdir(parents=True, exist_ok=True)
    for k, v in views.items():
        v.to_csv(INTERIM / f"view_{k}.csv", index=False)
    s = summarize(views)
    s.to_csv(TAB / "cleaning_log.csv", index=False)
    print(s.to_string(index=False))
```

src/stats_utils.py

```
"""Association statistics with effect sizes (never report a p-value alone)."""
import numpy as np
import pandas as pd
from scipy import stats
from statsmodels.stats.multitest import multipletests
from config import TARGET, SEED


def cramers_v(x, y) -> float:
    chi2 = stats.chi2_contingency(pd.crosstab(x, y), correction=False)[0]
    return float(np.sqrt(chi2 / len(x)))


def odds_ratio_ci(x_bin, y_bin, z=1.96):
    """Odds ratio (exposed vs unexposed) with Wald 95% CI. Inputs are 0/1 arrays."""
    x, y = np.asarray(x_bin), np.asarray(y_bin)
    a, b = ((x == 1) & (y == 1)).sum(), ((x == 1) & (y == 0)).sum()
    c, d = ((x == 0) & (y == 1)).sum(), ((x == 0) & (y == 0)).sum()
    orr = (a * d) / (b * c)
    se = np.sqrt(1 / a + 1 / b + 1 / c + 1 / d)
    return float(orr), float(np.exp(np.log(orr) - z * se)), float(np.exp(np.log(orr) + z * se))


def permutation_p(x_bin, y_bin, n_perm=5_000, seed=SEED) -> float:
    """Two-sided permutation p-value for the difference in positive rates between x=1 and x=0."""
    rng = np.random.default_rng(seed)
    x, y = np.asarray(x_bin), np.asarray(y_bin)
    obs = y[x == 1].mean() - y[x == 0].mean()
    hits = sum(abs((yp := rng.permutation(y))[x == 1].mean() - yp[x == 0].mean()) >= abs(obs) for _ in range(n_perm))
    return (hits + 1) / (n_perm + 1)


def feature_family(df: pd.DataFrame, features) -> pd.DataFrame:
    """Chi-square + Cramer's V for each binary feature; Benjamini-Hochberg correction across the family."""
    rows = []
    for c in features:
        chi2, p, _, _ = stats.chi2_contingency(pd.crosstab(df[c], df[TARGET]), correction=False)
        rows.append((c, float(np.sqrt(chi2 / len(df))), float(p)))
    out = pd.DataFrame(rows, columns=["feature", "cramers_v", "p"])
    out["q_BH"] = multipletests(out["p"], method="fdr_bh")[1]
    return out.sort_values("p").reset_index(drop=True)


def global_lr_test(df: pd.DataFrame, features) -> dict:
    """Likelihood-ratio test of ALL features jointly vs intercept-only logit (inference, not a predictive model)."""
    import statsmodels.api as sm
    m = sm.Logit(df[TARGET], sm.add_constant(df[features].astype(float))).fit(disp=0)
    return {"lr_chi2": float(m.llr), "df": int(m.df_model), "p": float(m.llr_pvalue), "pseudo_r2": float(m.prsquared)}
```

src/features.py

```
"""Feature engineering with a written specification. Scalers are FIT on training data only."""
import numpy as np
import pandas as pd
from config import SYMPTOMS_RESP, SYMPTOMS_OTHER, ID
from data_io import BINARY_12


def recode_binary(df: pd.DataFrame) -> pd.DataFrame:
    out = df.copy()
    for c in BINARY_12:
        out[c] = (out[c] == 2).astype("int8")                  # 1/2 coding -> 0/1 (2 = YES)
    return out.rename(columns={"GENDER": "GENDER_CODE"})       # sex mapping unverified: never call it male/female


def fit_age_scaler(train_age: pd.Series) -> dict:
    return {"mean": float(train_age.mean()), "std": float(train_age.std(ddof=0))}


def add_features(df: pd.DataFrame, age_scaler: dict) -> pd.DataFrame:
    out = recode_binary(df)
    out["AGE_Z"] = (out["AGE"] - age_scaler["mean"]) / age_scaler["std"]
    out["AGE_BAND"] = pd.cut(out["AGE"], [0, 49, 59, 69, 79, 120], labels=["<50", "50-59", "60-69", "70-79", "80+"])
    out["IN_SCREEN_AGE"] = out["AGE"].between(50, 80).astype("int8")                          # USPSTF 2021 age range
    out["RESP_SYMPTOM_COUNT"] = out[SYMPTOMS_RESP].sum(axis=1).astype("int8")                 # 0-4
    out["SYMPTOM_COUNT_ALL"] = out[SYMPTOMS_RESP + SYMPTOMS_OTHER].sum(axis=1).astype("int8")  # 0-6
    out["SMOKER_IN_SCREEN_AGE"] = (out["SMOKING"] & out["IN_SCREEN_AGE"]).astype("int8")      # eligibility PROXY only
    return out.drop(columns=[ID], errors="ignore")                                            # identifier never a feature


FEATURE_SETS = {
    "F0_raw15": ["GENDER_CODE", "AGE"] + BINARY_12,
    "F1_raw_plus_engineered": ["GENDER_CODE", "AGE_Z"] + BINARY_12
    + ["IN_SCREEN_AGE", "RESP_SYMPTOM_COUNT", "SYMPTOM_COUNT_ALL", "SMOKER_IN_SCREEN_AGE"],
    "F2_prior_driven": ["AGE_Z", "SMOKING", "IN_SCREEN_AGE", "RESP_SYMPTOM_COUNT", "SMOKER_IN_SCREEN_AGE"],
}


def verify(fe: pd.DataFrame) -> None:
    assert ID not in fe.columns, "identifier leaked into features"
    assert fe["RESP_SYMPTOM_COUNT"].between(0, 4).all() and fe["SYMPTOM_COUNT_ALL"].between(0, 6).all()
    assert set(fe["SMOKING"]) <= {0, 1} and fe.isna().sum().sum() == 0
    assert (fe["SYMPTOM_COUNT_ALL"] >= fe["RESP_SYMPTOM_COUNT"]).all()
    assert set(np.unique(fe["SMOKER_IN_SCREEN_AGE"])) <= {0, 1}
    for name, cols in FEATURE_SETS.items():
        assert all(c in fe.columns for c in cols), f"{name} references a missing column"
```

tests/test_pipeline.py

```
import sys
import pathlib
sys.path.insert(0, str(pathlib.Path(__file__).resolve().parents[1] / "src"))
import pytest
from config import TARGET, ID
from data_io import load_raw, FEATURES
from audit import leading_block_end, duplicate_report, auc_of_id
from clean import make_views
from features import add_features, fit_age_scaler, verify, FEATURE_SETS


@pytest.fixture(scope="module")
def df():
    return load_raw()                       # also verifies SHA-256 and schema


def test_raw_shape(df):
    assert df.shape == (22457, 17)


def test_leading_block(df):
    fp = leading_block_end(df)
    assert fp == 1455 and df.iloc[:fp][TARGET].eq(0).all()
    assert len(df.iloc[:fp][FEATURES].drop_duplicates()) == 103


def test_duplicate_and_conflict_counts(df):
    r = duplicate_report(df)
    assert r["dup_rows_features_and_label"] == 2573 and r["conflicting_patterns"] == 133


def test_id_is_a_leak(df):
    assert 0.65 < auc_of_id(df) < 0.69     # identifier alone separates classes -> must never be a feature


def test_views(df):
    v = make_views(df)
    assert [len(x) for x in v.values()] == [22457, 19884, 19618, 19781]
    assert not v["B_dedup"].duplicated(subset=FEATURES + [TARGET]).any()
    assert v["D_dedup_nolead"].index.min() >= 1455


def test_features_no_leakage(df):
    B = make_views(df)["B_dedup"]
    fe = add_features(B, fit_age_scaler(B.AGE))
    verify(fe)
    assert ID not in sum(FEATURE_SETS.values(), []) and TARGET not in sum(FEATURE_SETS.values(), [])
    assert abs(fe["AGE_Z"].mean()) < 1e-9
```

Makefile

```
export PYTHONPATH := src
.PHONY: audit clean test all
audit:
    python src/audit.py
clean:
    python src/clean.py
test:
    pytest -q
all: audit clean test
```

Expected: make all

```
$ make all            # export PYTHONPATH=src is set inside the Makefile
python src/audit.py   -> first_positive_index 1455, leading_block_distinct_patterns 103, auc_patient_id_alone 0.67,
                         dup_rows_features_and_label 2573, conflicting_patterns 133
python src/clean.py   -> A_raw 22457 (0.8209) | B_dedup 19884 (0.8661) | C_dedup_noconflict 19618 (0.8711) | D_dedup_nolead 19781 (0.8706)
pytest -q             -> 6 passed
```

Appendix C. Submission-day checklist

Done	Item	Owner
[ ]	Repository at tag v0.1-midterm; README lists the 5 commands; requirements.txt pinned	A, D
[ ]	Clean-clone run: make all in under 5 minutes; pytest green; log saved	D
[ ]	results.json regenerated; consistency script reports zero mismatches between code, report and slides	G
[ ]	Report PDF: six sections, numbered figures and tables with captions, references, reproducibility appendix	G
[ ]	Slides PDF and PPTX: 14 slides plus backup slides, footer disclaimer, slide numbers, timing sheet printed	A
[ ]	No clinical-claim phrases (search the banned list in the report and slides)	A
[ ]	Data access: CHECKSUMS.txt and source link, or the CSV if the licence permits	B
[ ]	Two timed rehearsals at or under 19:30; two Q and A rounds logged; every member at 80% or more	A
[ ]	Backups: slides on USB and email, notebook HTML exports, laptop charger and adapter, a second presenter laptop	all
[ ]	Roles for the Q and A: who answers what first; rule for deferring questions to the right member	A

Appendix D. Templates

Decision record (docs/decision_log.md)

```
ID: D-xx        Date: 2026-10-xx        Proposed by:        Decided by:
Context (what forced the decision, with numbers):
Options considered (1-3 lines each):
Decision:
Consequences / what we give up:
Revisit when:
```

Daily standup (docs/standup.md)

```
Date: 2026-10-xx
Name | Done since last standup (with evidence) | Doing next (task ID, due) | Blocked by (who/what)
Integration status: main green? yes/no   Numbers changed in results.json? list
Decisions needed from A:
```

Per-slide script card

```
Slide #  Speaker  Time  | One-sentence message | Figure axes to describe first | Numbers to say (from results.json)
Why this method (one sentence) | Limitation to mention | Handover sentence to next speaker
```

Appendix E. Open questions for the instructor and data provider

ID	Question	Why it matters	Interim assumption
Q1	Where did the ZIP come from (link, instructor, team member)? Is there a data dictionary and licence?	Provenance and redistribution; label meaning	Unknown; do not redistribute without checking
Q2	Is this dataset fixed for the whole project, or may we add or switch data?	Determines the semester track (Ch. 14)	Fixed for the midterm
Q3	Report and code submission deadline, format, page limit, language?	Schedule and structure	Before the presentation, PDF, 10-14 pages, English
Q4	Length and format of the Q and A; are questions individual?	Cross-training intensity	5-10 minutes, individual
Q5	Is it acceptable to present a data-integrity finding as the main result, and to use cleaned views?	Mitigates risk R3	Yes, with sensitivity analysis
Q6	What does LUNG_CANCER = 1 mean and what does GENDER = 1 mean?	Label validity; subgroup wording	Recorded label; GENDER_CODE
Q7	Will the final deliverable require modelling on this file?	Plan in Ch. 14	Yes, with leakage-safe protocol
Q8	Presentation language and room/technology details?	Delivery	English; own laptop plus PDF backup
Q9	Team availability per person per day (hours)?	Task feasibility	At least 3 hours per day each

Appendix F. Glossary

Term	Meaning
Association	Statistical dependence between two variables; not causation.
Benjamini-Hochberg (BH)	Procedure that controls the false discovery rate across many tests.
Calibration	Agreement between predicted risks and observed frequencies.
Cramer's V	Effect size 0 to 1 for chi-square tests.
CLES	Common-language effect size: P(random positive exceeds random negative).
Confounding	A third factor linked to both variables that creates or hides an association.
Construct validity	Whether a variable measures what it claims to.
CRISP-DM	Cross-Industry Standard Process for Data Mining: six phases with feedback loops.
Data leakage	Information in training that would not be legitimately available or independent at test time.
Effect size	Magnitude of an association independent of sample size.
FDR	False discovery rate: expected share of false positives among rejected nulls.
Grouped CV	Cross-validation that keeps all rows of a group (here identical patterns) in the same fold.
IQR rule	Flags values beyond Q1 - 1.5 IQR or Q3 + 1.5 IQR.
MCAR/MAR/MNAR	Missing completely at random / at random given observed data / not at random.
Odds ratio	Ratio of odds of the outcome between exposed and unexposed.
Pack-year	One pack (20 cigarettes) per day for one year.
Permutation test	Significance from shuffling labels to build the null distribution.
Power	Probability of detecting a true effect of a given size.
PR-AUC	Area under the precision-recall curve; chance level equals prevalence.
Pseudo-replication	Treating copies of one observation as independent evidence.
Research prototype	Work for study, not validated for clinical use.
Sentinel value	A code such as 999 used to mark missing data.
TRIPOD+AI	27-item reporting guideline for prediction-model studies (BMJ 2024).
VIF	Variance inflation factor: multicollinearity measure; 1 means none.
View (A-D)	One of four pre-specified analysis datasets differing in cleaning policy.